Student Enrollment & Course Management System (SECMS)

# Project Documentation


## TEAM MEMBERS:

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## 1.INTRODUCTION

### 1.1Project Overview:

This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress. The solution also enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management.

With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can ensure that all enrollment operations—from admission to course completion—are handled systematically and with continuous monitoring of key performance indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

### 1.2 Purpose:

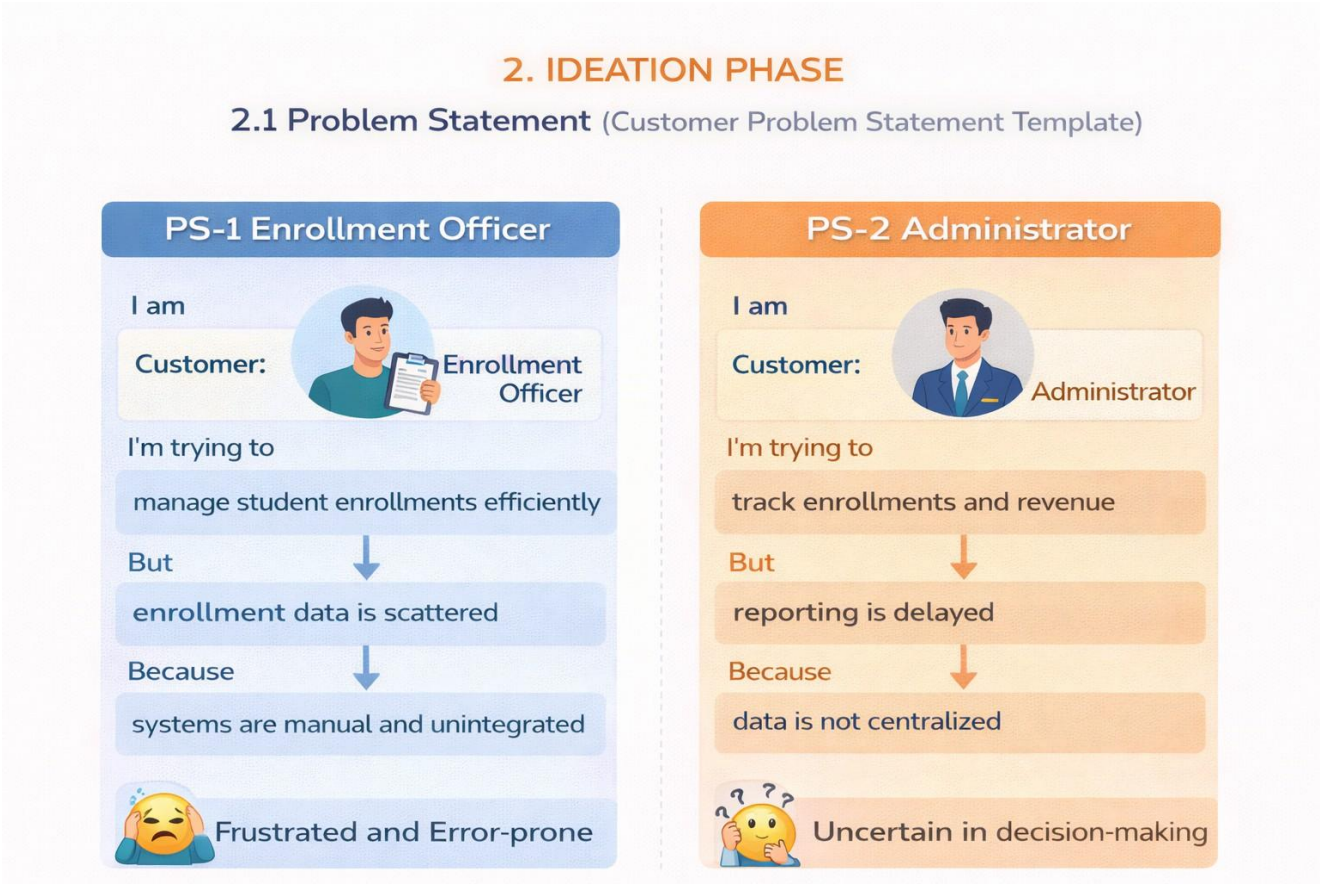
The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:

Accurate student data management
Structured enrollment workflows
Automated approvals and validations
Transparent reporting and monitoring

## 1. IDEATION PHASE

### 1.1 Problem Statement (Customer Problem Statement Template)

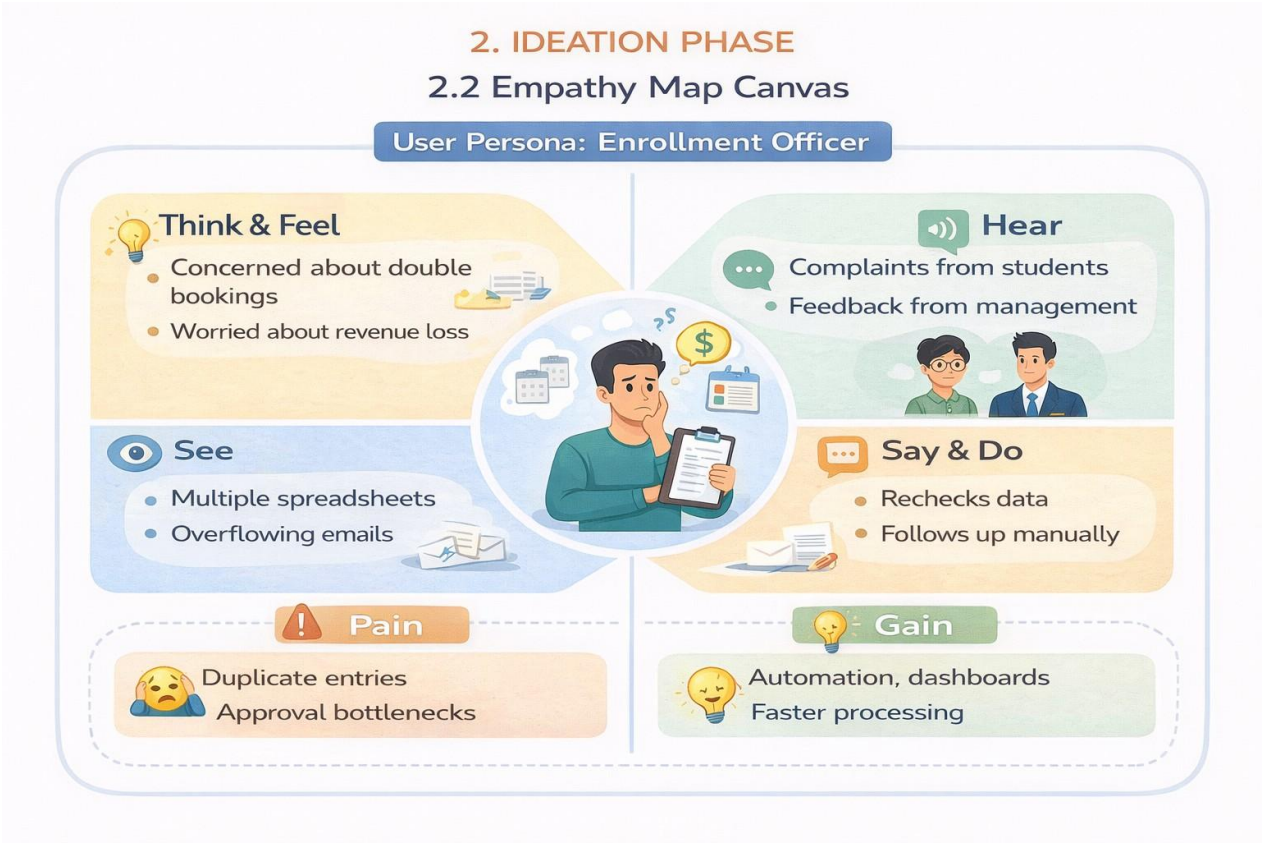
| PS | I am (Customer) | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| PS-1 | Enrollment Officer | manage student enrollments efficiently | enrollment data is scattered | systems are manual and unintegrated | frustrated and error-prone |
| PS-2 | Administrator | track enrollments and revenue | reporting is delayed | data is not centralized | uncertain in decision-ma king |




## 1.2 Empathy Map Canvas

**User Persona:** Enrollment Officer

- **Think & Feel:** Concerned about double bookings and revenue loss
- **Hear:** Complaints from students and management
- **See:** Multiple spreadsheets and emails
- **Say & Do:** Rechecks data, follows up manually
- **Pain:** Duplicate entries, approval bottlenecks
- **Gain:** Automation, dashboards, faster processing




## 1.3 Brainstorming & Idea Prioritization

### 1.3.1. Brainstormed Ideas

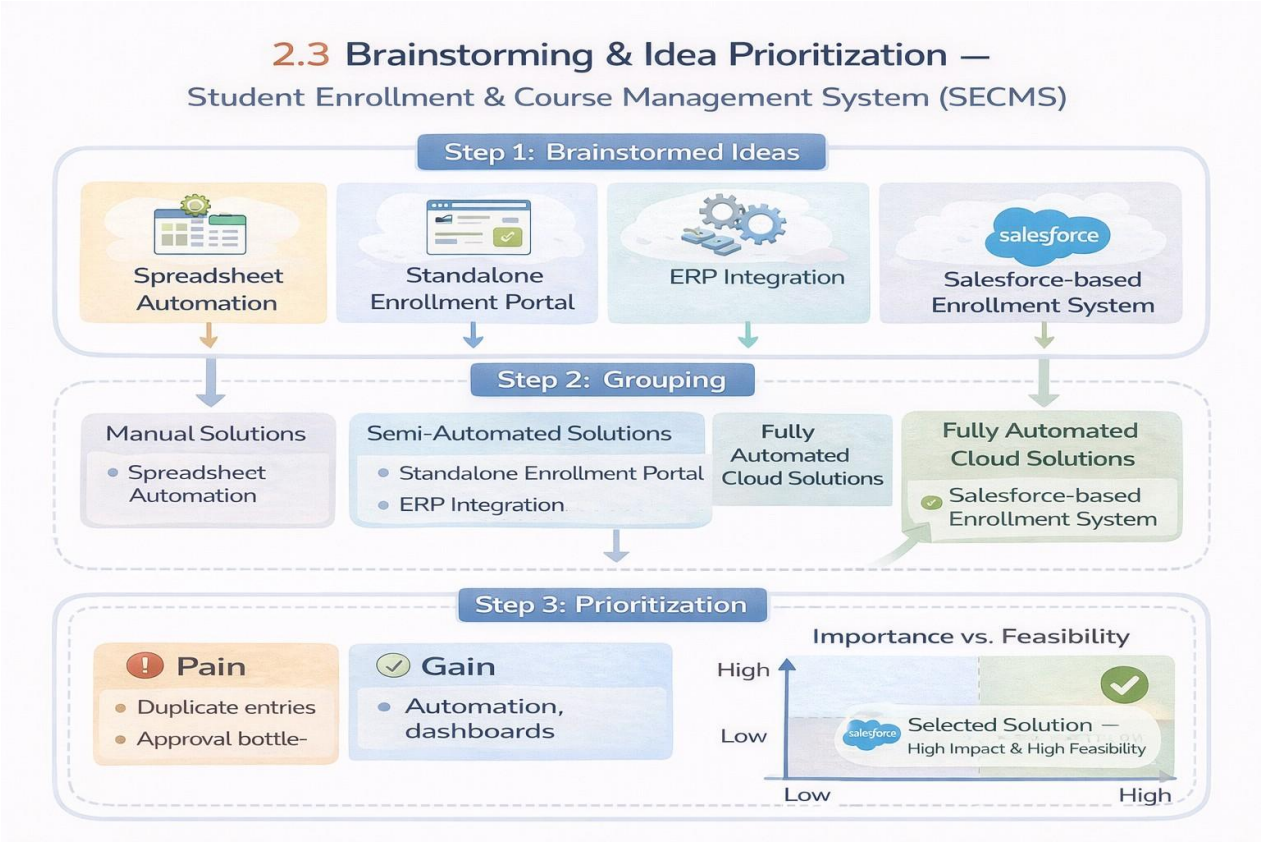
- Spreadsheet automation
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

### 1.3.2. Grouping

- Manual solutions
- Semi-automated solutions
- Fully automated cloud solutions

### 1.3.3. Prioritization

Salesforce-based SECMS was selected due to **high feasibility** and **high impact**.

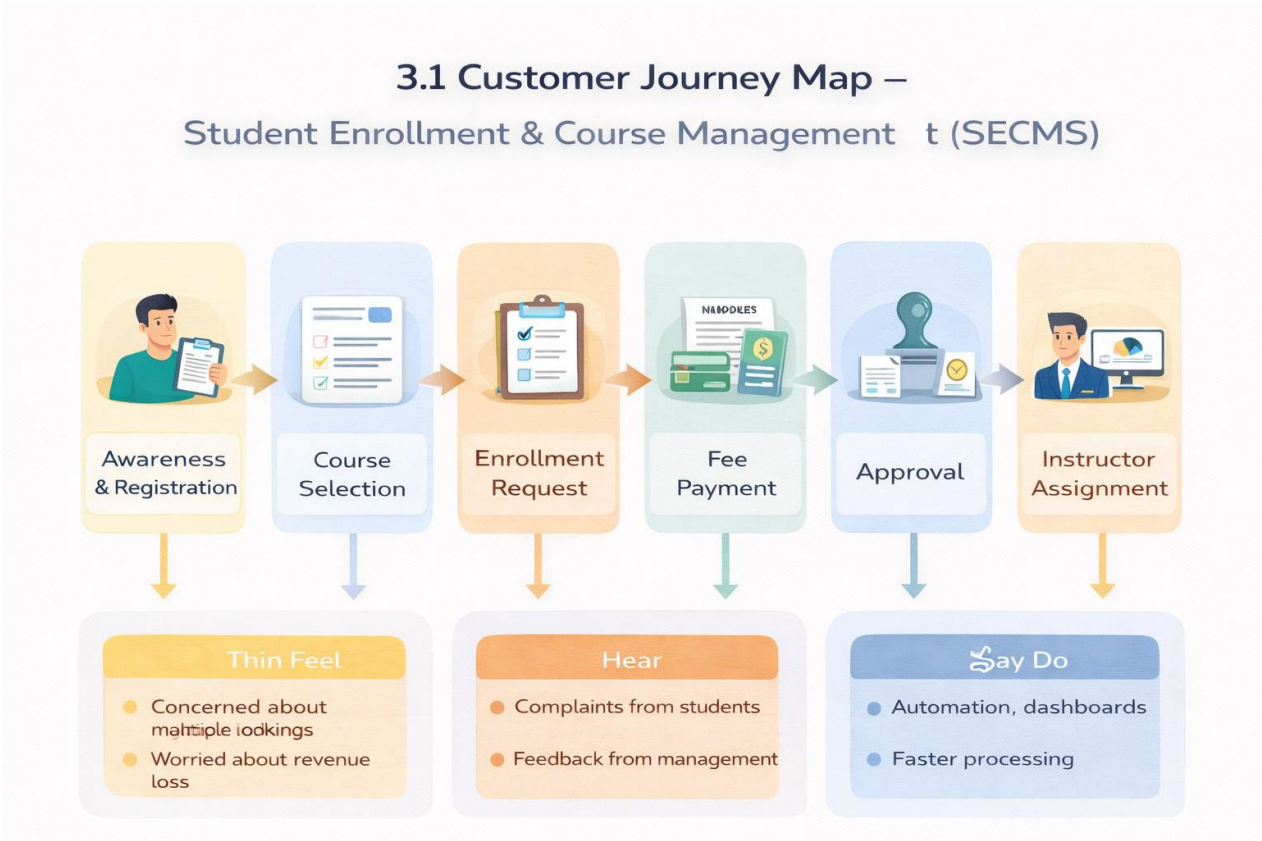



## 4. REQUIREMENT ANALYSIS PHASE

### 1.4 Customer Journey Map

Stages covered:

- Awareness & Registration
- Course Selection
- Enrollment Request
- Fee Payment
- Approval
- Instructor Assignment




## 1.5 Solution Requirements:

### Functional Requirements

| FR No | Epic | Description |
|---|---|---|
| FR-1 | Student Management | Manage student records |
| FR-2 | Course Management | Create and manage courses |
| FR-3 | Enrollment Processing | Request, approve, reject enrollments |
| FR-4 | Instructor Assignment | Auto-assign instructors |

| | | |
|---|---|---|
| FR-5 | Reporting | Dashboards and reports |

**Non-Functional Requirements**

**NFR-1: Usability**

The system shall provide a simple, intuitive, and user-friendly interface that enables enrollment officers, administrators, instructors, and students to perform operations efficiently with minimal training. Navigation shall be organized through clearly labeled tabs, dashboards, and forms to reduce user effort and errors.

**Examples:**

- Quick student registration form
- One-click enrollment approval
- Role-based dashboards

**NFR-2: Security**

The system shall ensure the confidentiality, integrity, and protection of student and institutional data through secure authentication, authorization, and access controls. Sensitive information such as personal data, enrollment history, and academic records shall only be accessible to authorized users.

**Examples:**

- Only admins can modify course details
- Instructors view only assigned students
- Students cannot access other student records

**NFR-3: Reliability**

The system shall operate consistently and accurately without failures during enrollment processing, approvals, and reporting. Automation logic shall execute correctly to avoid duplicate or incorrect enrollments.

## Examples:

- No duplicate course registration
- Accurate status updates after approvals
- Consistent report generation

### NFR-4: Performance

The system shall provide fast response times for all operations to ensure smooth and uninterrupted user experience, especially during peak admission periods.

### Examples:

- Quick dashboard refresh
- Fast search for student records
- No lag during bulk uploads

### NFR-5: Scalability

The system shall support future growth in the number of students, courses, departments, and users without degradation in performance or functionality. It shall allow easy addition of new modules and features.

### Examples:

- Add new departments without redesign
- Handle increased admissions annually
- Extend to mobile or portal integration

### NFR-6: Availability

The system shall be continuously accessible to users during institutional working hours and must minimize downtime. The platform shall support high availability for academic operations.

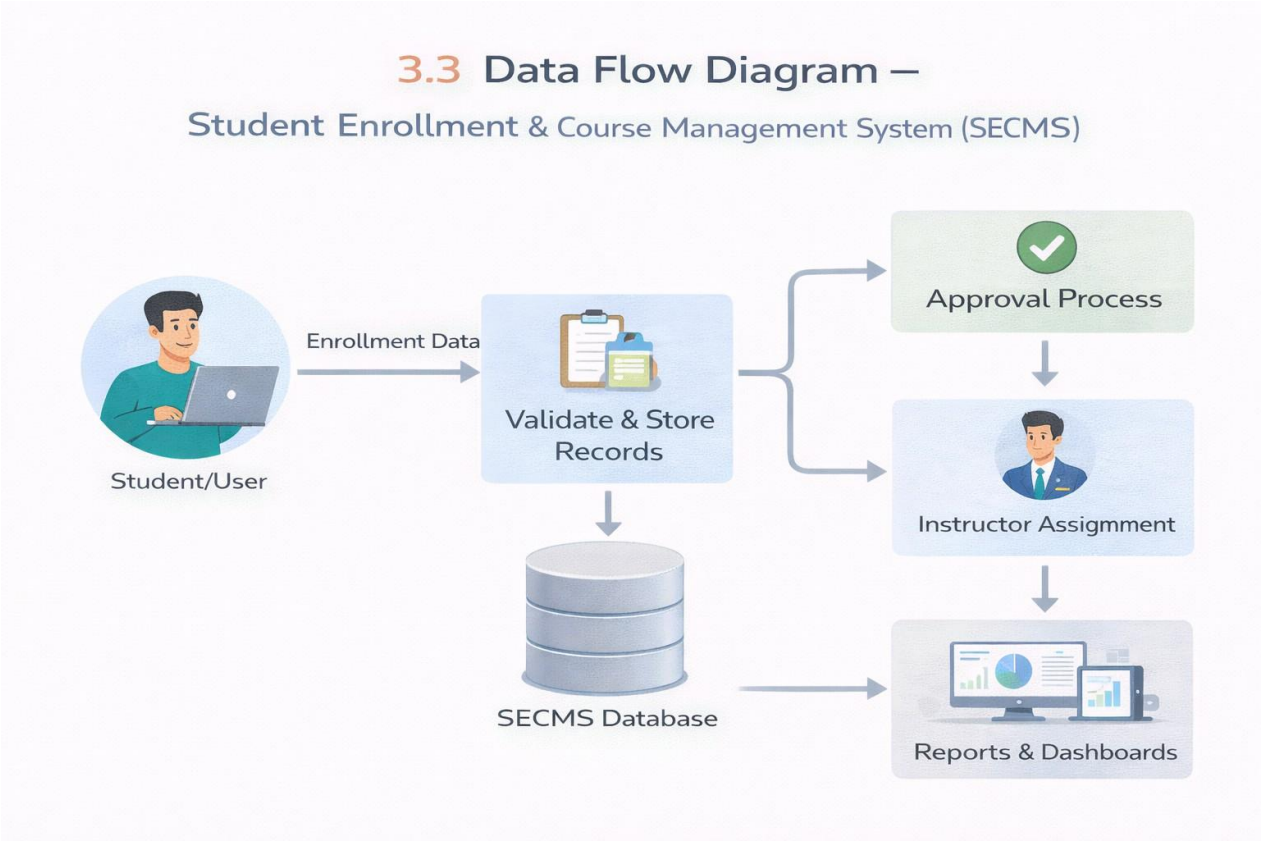
### Examples:

- 24/7 cloud access
- Reliable during peak admission season
- Automatic backup of student records

## 1.6 Data Flow Diagram:

- User inputs student and enrollment data
- System validates and stores records
- Approval process routes data
- Reports and dashboards consume processed data.

The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.




## 1.7 Technology Stack

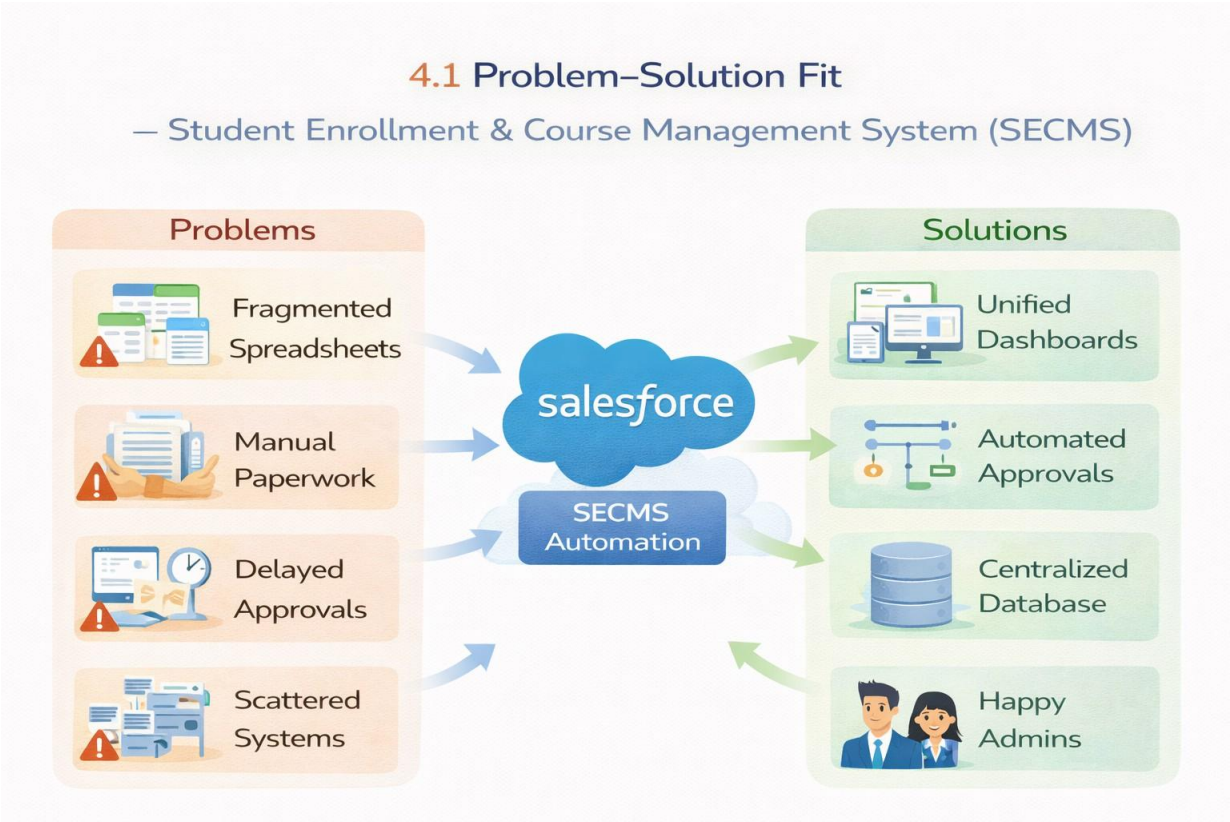
| Layer | Technology |
|---|---|
| UI | Salesforce Lightning |

| | |
|---|---|
| Logic | Salesforce Flows |
| Database | Salesforce Objects |
| Automation | Validation Rules, Approval Process |
| Reporting | Reports & Dashboards |

# 2. PROJECT DESIGN PHASE

## 2.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.



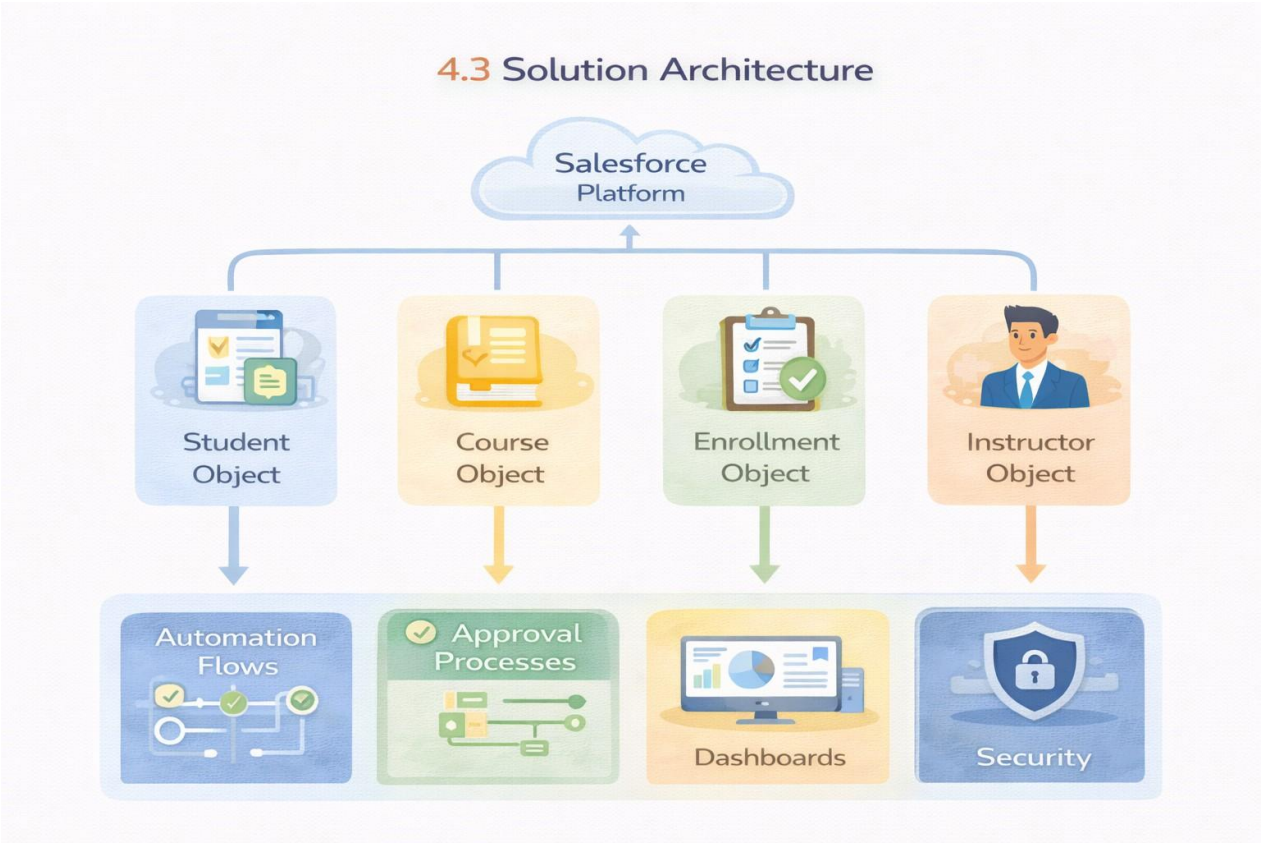
## 2.2 Proposed Solution

### Functional Requirements

| Parameter | Description |
|---|---|
| Problem | Manual enrollment handling |
| Solution | Salesforce-based SECMS |
| Novelty | End-to-end automation |
| Social Impact | Improved student experience |
| Revenue Model | Institutional deployment |
| Scalability | Multi-department support |

## 2.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers.




# 3. PROJECT PLANNING & SCHEDULING

## 3.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

**Product Backlog, Sprint Schedule, and Estimation (4 Marks)**

Use the below template to create product backlog and sprint schedule

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority | Team Members |
|---|---|---|---|---|---|---|
| Sprint-1 | Developer Setup | USN-1 | As a system administrator, I want to create a Salesforce developer account so that I can configure and deploy the Student Enrollment system.. | 1 | High | Member1 |
| Sprint-1 | Data Modeling | USN-2 | As an admin, I want to create custom objects for Student, | 3 | High | Member2 |

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority | Team Members |
|---|---|---|---|---|---|---|
| | | | Course, Enrollment, and Instructor so that academic data can be stored and managed efficiently. | | | |
| Sprint-2 | Data Modeling | USN-3 | As a user, I want dedicated tabs for each object so that I can easily navigate between students, courses, enrollments, and instructors. | 3 | High | Member3 |
| Sprint-2 | Data Modeling | USN-4 | As an admin, I want to create relevant fields such as student details, course information, and enrollment status so that complete records can be captured accurately. | 5 | High | Member4 |
| Sprint-2 | Data Modeling | USN-5 | As an enrollment officer, I want a centralized Lightning App so that I can access all academic operations from one interface. | 3 | High | Member4 |
| Sprint-2 | Data Modeling | USN-6 | As an admin, I want to define relationships between students, courses, and enrollments so that the system can logically link and track academic activities. | 5 | Medium | Member1 |
| Sprint-2 | Data Modeling | USN-7 | As a user, I want customized page layouts so that only relevant information is displayed, improving clarity and ease of data entry. | 5 | High | Member2 |
| Sprint-3 | Automation | USN-8 | As a system, I want to validate input data such as mandatory fields and enrollment limits so that incorrect or incomplete records are prevented. | 3 | High | Member4 |
| Sprint-3 | Automation | USN-9 | As a system, I want automated flows to handle enrollment processing and notifications so that manual effort is reduced and processes become faster. | 3 | High | Member4 |
| Sprint-3 | Automation | USN-10 | As an administrator, I want an approval process for enrollment requests so that registrations are verified before confirmation. | 5 | High | Member1 |
| Sprint-4 | Security | USN-11 | As an admin, I want to configure role-based security so that users can only access data relevant to their responsibilities. | 5 | High | Member2 |
| Sprint-5 | Reports | USN-12 | As a manager, I want reports on student enrollments, course registrations, and instructor assignments so that I can monitor academic operations. | 4 | High | Member3 |
| Sprint-6 | Dashboards | USN-13 | As a manager, I want visual dashboards showing enrollment | 4 | Medium | Member4 |

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority | Team Members |
|---|---|---|---|---|---|---|
| | | | statistics and trends so that I can quickly analyze system performance. | | | |

**Project Tracker, Velocity & Burndown Chart: (4 Marks)**

| Sprint | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|---|---|---|---|---|---|---|
| Sprint-1 | 20 | 6 Days | 24 Oct 2022 | 29 Oct 2022 | 20 | 29 Oct 2022 |
| Sprint-2 | 20 | 6 Days | 31 Oct 2022 | 05 Nov 2022 | | |
| Sprint-3 | 20 | 6 Days | 07 Nov 2022 | 12 Nov 2022 | | |
| Sprint-4 | 20 | 6 Days | 14 Nov 2022 | 19 Nov 2022 | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |

## 4. Project Development Phase:

## Project Flow:

Milestone 1: Salesforce Developer Account Creation
Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)
Milestone 3: Tabs Creation
Milestone 4: Lightning App Creation
Milestone 5: Fields Creation
Milestone 6: Create Relationships

Milestone 7: Page Layout Customization
Milestone 8: Validation Rules
Milestone 9: Flows (Automation)
Milestone 10: Approval Process
Milestone 11: Reports
Milestone 12: Dashboards
Milestone 13: Security Setup
Milestone 14: Conclusion

**What you'll learn**

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows
7. Developing reports & dashboards
8. Setting up security with profiles, permission sets, and sharing
9. End-to-end application testing

## Milestone 1-Salesforce :

## Introduction:

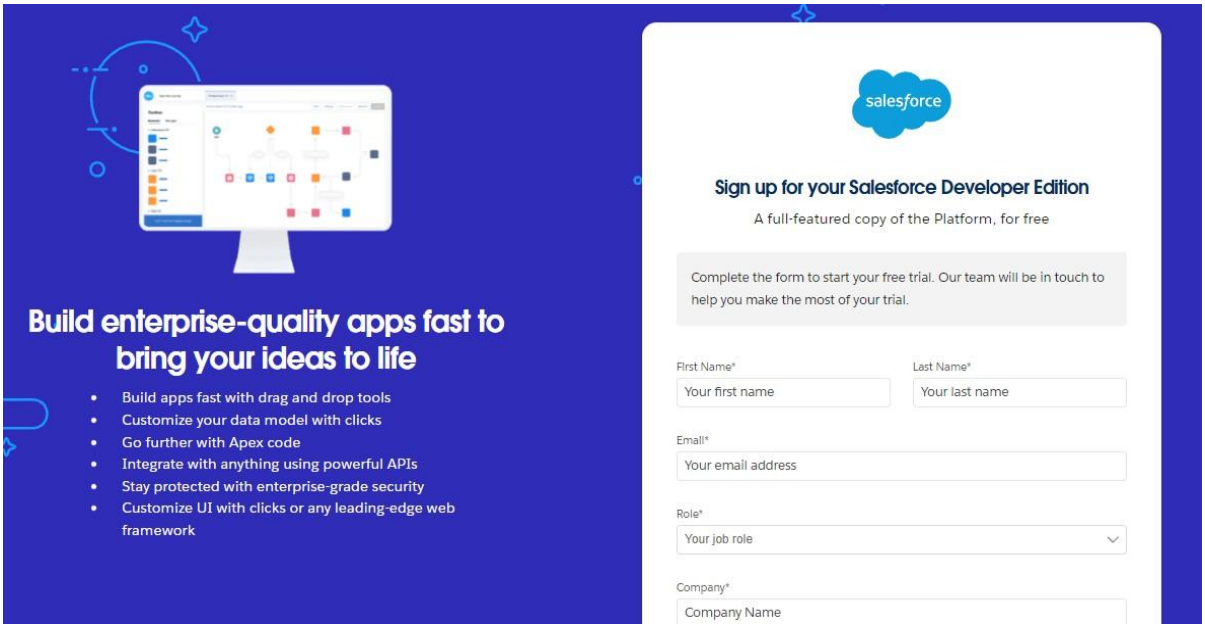
Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don't know where you should start on your learning journey? If you've answered yes to any of these questions, then you're in the right place. This module is for you.
Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?".

## What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud. So what does that really mean? Well, before Salesforce, your contacts, emails, follow-up tasks, and prospective deals might have been organized something like this:
https://youtu.be/r9EX3lGde5k

## Activity 1: Creating Developer Account:

Creating a developer org in salesforce.
Go to https://developer.salesforce.com/signup On the sign up form, enter the following details :



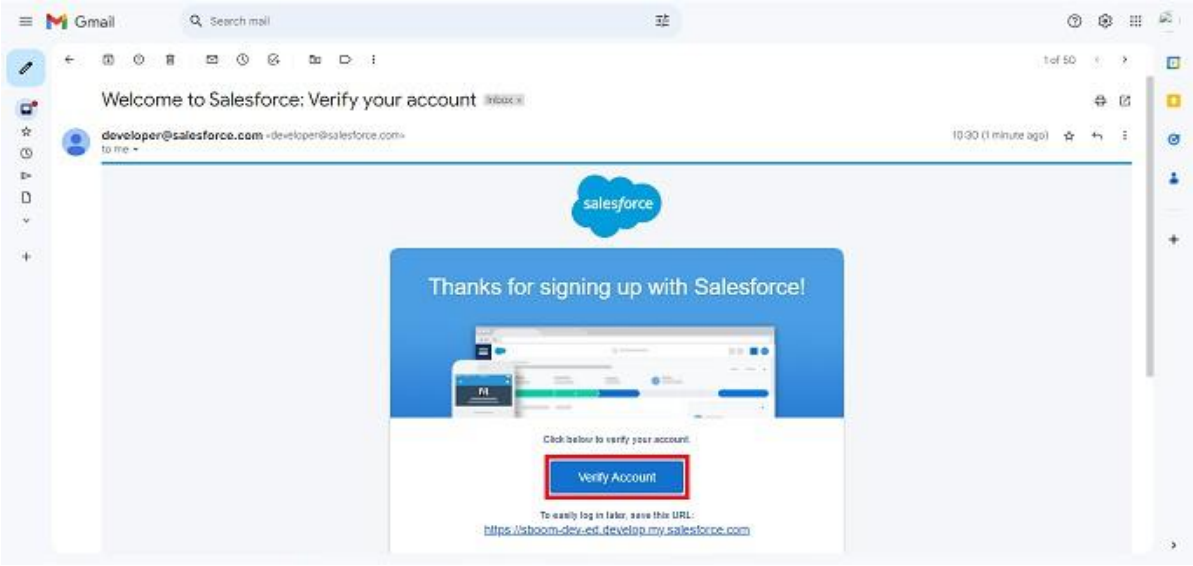

First name & Last name

Email
Role : Developer

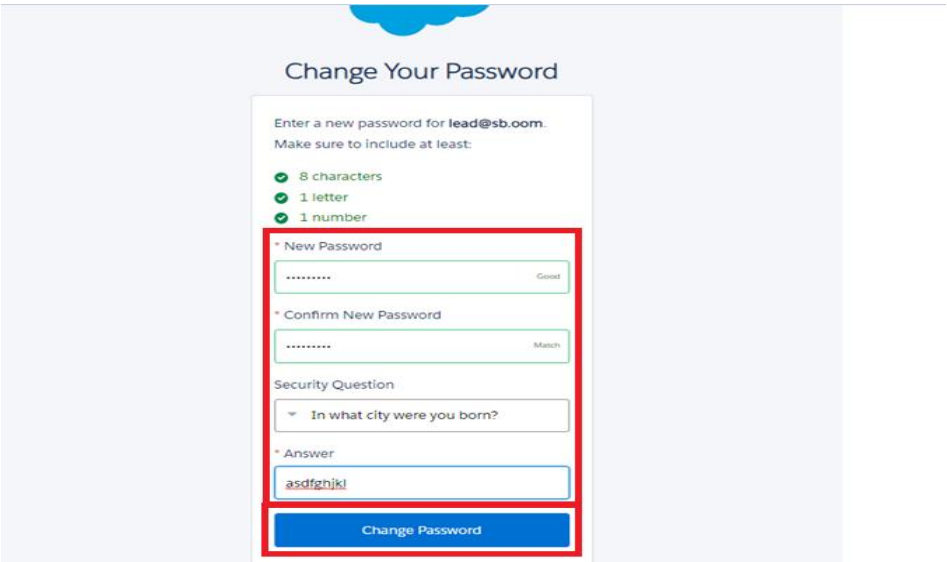
Company: College Name County: India
Postal Code: pin code
Username: should be a combination of your name and company
This need not be an actual email id, you can give anything in the format:
[username@organization.com](mailto:username@organization.com)
Click on sign me up after filling these.

## **Activity 2: Account Activation**:

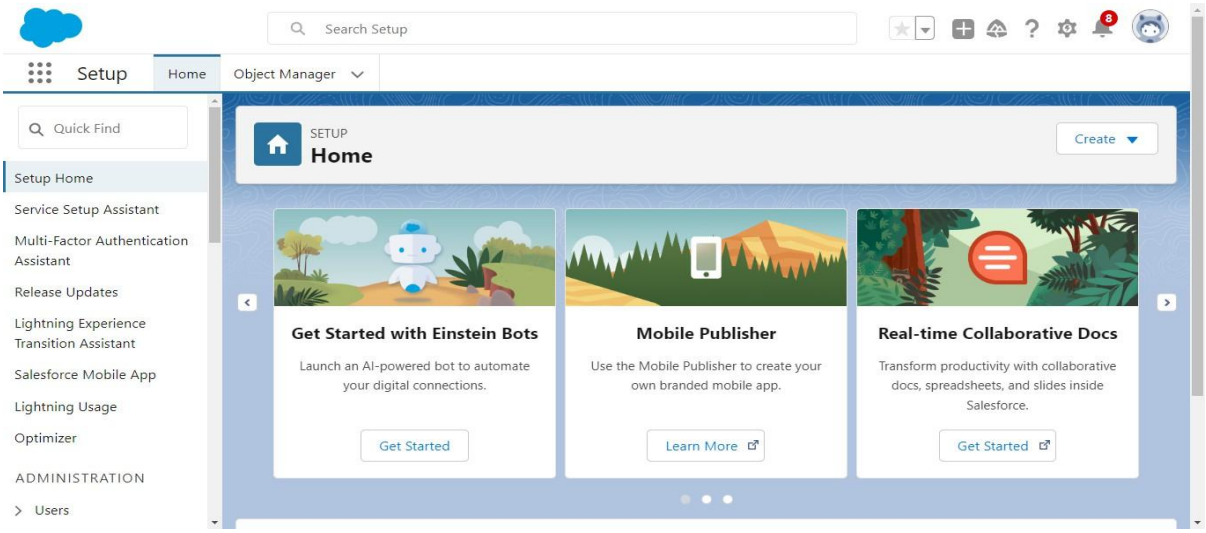
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.




2. Click on Verify Account
3. Give a password and answer a security question and click on change password.




4. Then you will redirect to your salesforce setup page.



# Milestone 2 – Objects:

## Activity 1: Creating Student  Object:

The purpose of creating the Student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.

To create the Student object:

Go to the Setup page. Click on

Object Manager.

Click on Create → Custom Object.

Enter the details as below:

1. Label: Student
2. Plural Label: Students
3. Object Name: Student
4. Record Name: Student Name
5. Data Type: Text
6. Under Optional Features, select:
7. Allow Reports
8. Allow Activities (optional but recommended)
9. Track Field History (optional, if you want)
10. Under Search Status, select:
11. Allow Search
12. Click Save.

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

**Student**

- Details
- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules

**Custom Object Definition Edit** — Save | Save & New | Cancel

**Custom Object Information**

The singular and plural labels are used in tabs, page layouts, and reports.
**Be careful when changing the name or label as it may affect existing integrations and merge templates.**

Label: Student — **Example: Account**

Plural Label: Students — **Example: Accounts**

Starts with vowel sound: ☐

The Object Name is used when referencing the object via the API.

Object Name: Student — **Example: Account**

Description:

Context-Sensitive Help Setting:
- (●) Open the standard Salesforce.com Help & Training window
- ( ) Open a window using a Visualforce page

Content Name: --None--

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

**Student**

- Details
- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules

**Enter Record Name Label and Format**

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Cas
called "Name" when referenced via the API.

Record Name: Student Name — **Example: Account Name**

Data Type: Text — **Warning:** If you plan to insert a high volume of records in this object, via the API for example, use the Text data ty

**Optional Features**

- [x] Allow Reports
- [x] Allow Activities
- [x] Track Field History
- [ ] Allow in Chatter Groups
- [ ] Enable Licensing

**Object Classification**

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object.

- [x] Allow Sharing
- [x] Allow Bulk API Access
- [x] Allow Streaming API Access

**Deployment Status**

- ( ) In Development
- (●) Deployed

**Deployment Status**

- ( ) In Development
- (●) Deployed

**Search Status**

When this setting is enabled, your users can find records of this object type when they search. Learn more.

- [x] Allow Search

Save | Save & New | Cancel

In the same way Create **Courses, Instructors,** and **Enrolments objects**

## Milestone 3 - Fields:

### Table 1: Student Object Fields:

| Field Label | Data Type |
|---|---|
| Student Name | Text (Standard) |
| Email | Email |
| Phone | Phone |
| Date of Birth | Date |
| Student Status | Picklist |
| Address | Text Area |
| | |
| | |
| | |

| Field Label | Data Type |
|---|---|
| Course Name | Text (Standard) |
| Category | Picklist |
| Duration | Number/Text |
| Course Fee | Currency |
| Description | Text Area |

**Table 3: Instructor Object Fields:**

| Field Label | Data Type |
|---|---|
| Instructor Name | Text (Standard) |
| Instructor Code | Text |
| Expertise | Picklist |
| Phone | Phone |
| Email | Email |

**Table 4: Enrollment Object Fields:**

| Field Label | Data Type |
|---|---|
| Student | Lookup (Student) |
| Course | Lookup (Course) |
| Instructor | Lookup (Instructor) |
| Enrollment Date | Date |
| Enrollment Status | Picklist |

| | |
|---|---|
| Fees Paid | Checkbox |
| Total Amount | Currency |
| Comments | Text Area |

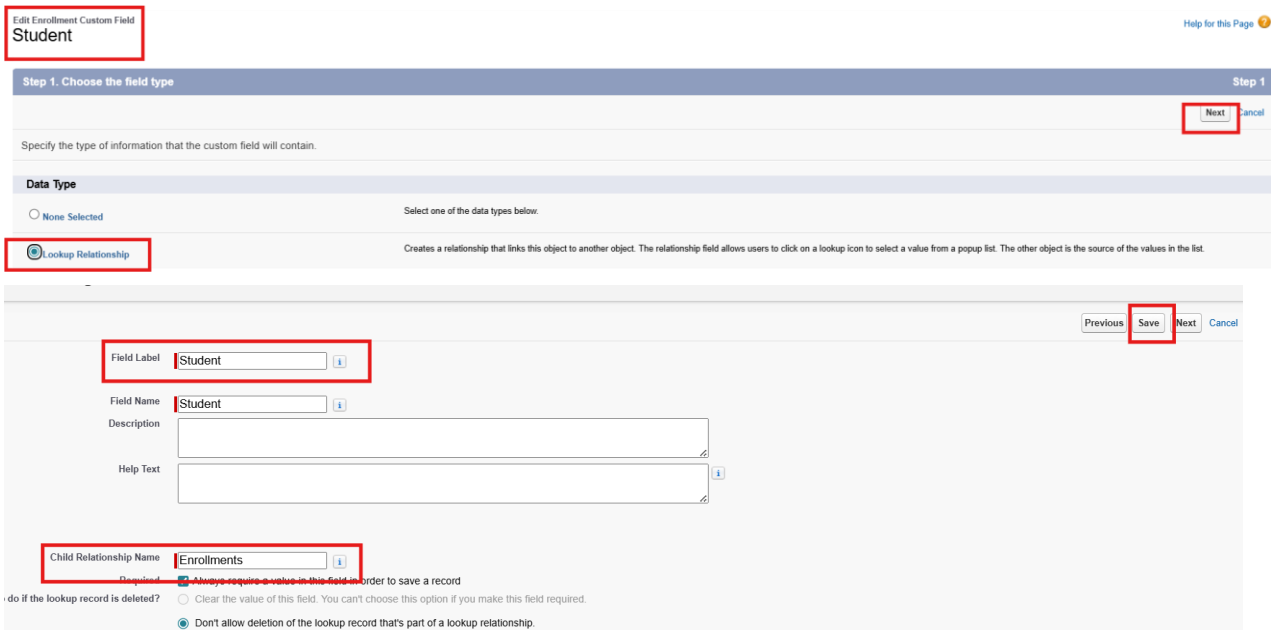
**Milestone 4 – Create Relationships:**

## 1. Activity 1: Create Lookup from Enrollment to Student:

1. Go to Setup → click Object Manager.
2. Type Enrollment in Quick Find → click Enrollment_____c object.
3. Click Fields & Relationships → click New.
4. Select Lookup Relationship → click Next.
5. For Related To, select Student___c →

**Enter Field Label: Student (API Name: Student_____c).**

1. Click Next through Field-Level Security and Layout options (ensure it's visible and added to layout).
2. Click Save.

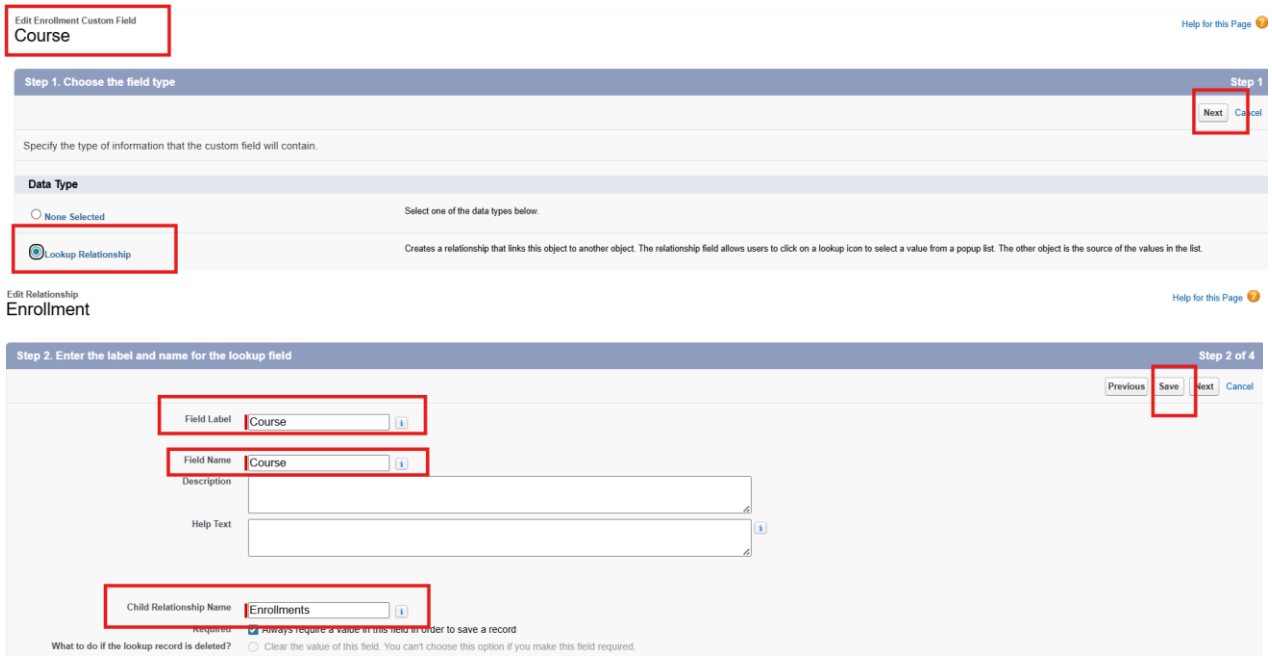



## Activity 2: Create Lookup from Enrollment to Course

Purpose:
To link each Enrollment to a Course.

Steps:

1. Go to Setup → Object Manager → Enrollment_____c.
2. Click Fields & Relationships → New.
3. Select Lookup Relationship → Next.
4. Related To: select Course_____c → click Next.
5. Field Label: Course (API Name: Course_____c).
6. Click Next (set visibility and layout) → Save.




## Activity 3: Create Lookup from Enrollment to Instructor:

Purpose:
To store which Instructor is assigned to a particular Enrollment.

Steps:

1. Go to Setup → Object Manager → Enrollment_____c.
2. Click Fields & Relationships → New.
3. Choose Lookup Relationship → Next.
4. Related To: select Instructor_____c → Next.
5. Field Label: Instructor (API Name: Instructor_____c).
6. Complete Field-Level Security and Page Layout options → Save

SETUP > OBJECT MANAGER

## Enrollment

- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits

Next Cancel

Specify the type of information that the custom field will contain.

**Data Type**

| | |
|---|---|
| None Selected | Select one of the data types below. |
| Auto Number | A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record. |
| Formula | A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change. |
| Roll-Up Summary | A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list. |
| Lookup Relationship | Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list. |

Enrollment

## New Relationship

Help for this Page

**Step 2. Choose the related object** — Step 2

Previous Next Cancel

Select the other object to which this object is related.

Related To: Instructor

Previous Next Cancel

**Step 3. Enter the label and name for the lookup field** — Step 3 of 6

Previous Next Cancel

Field Label: Instructor

Field Name: Instructor

Description:

Help Text:

Child Relationship Name: Enrollments

Required: ☑ Always require a value in this field in order to save a record

What to do if the lookup record is deleted?
- ○ Clear the value of this field. You can't choose this option if you make this field required.
- ◉ Don't allow deletion of the lookup record that's part of a lookup relationship.

Auto add to custom report type: ☑ Add this field to existing custom report types that contain this entity

Enrollment

## New Relationship

Help for this Page

**Step 6. Add custom related lists** — Step 6 of 6

Previous Save & New Save Cancel

Field Label: Instructor

Data Type: Lookup

Field Name: Instructor

Description:

Specify the title that the related list will have in all of the layouts associated with the parent.

Related List Label: Enrollments

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section of these page layouts. The field will not appear on any pages if you do not select a layout.

To change the location of this field on the page, you will need to customize the page layout. To change the values that appear, you will need to customize the Record Types.

| Add Related List | Page Layout Name |
|---|---|
| ☑ | Instructor Layout |

☑ Append related list to users' existing personal customizations

## Milestone 5 – Validation Rules:

### Activity 1: Student Email Domain Validation

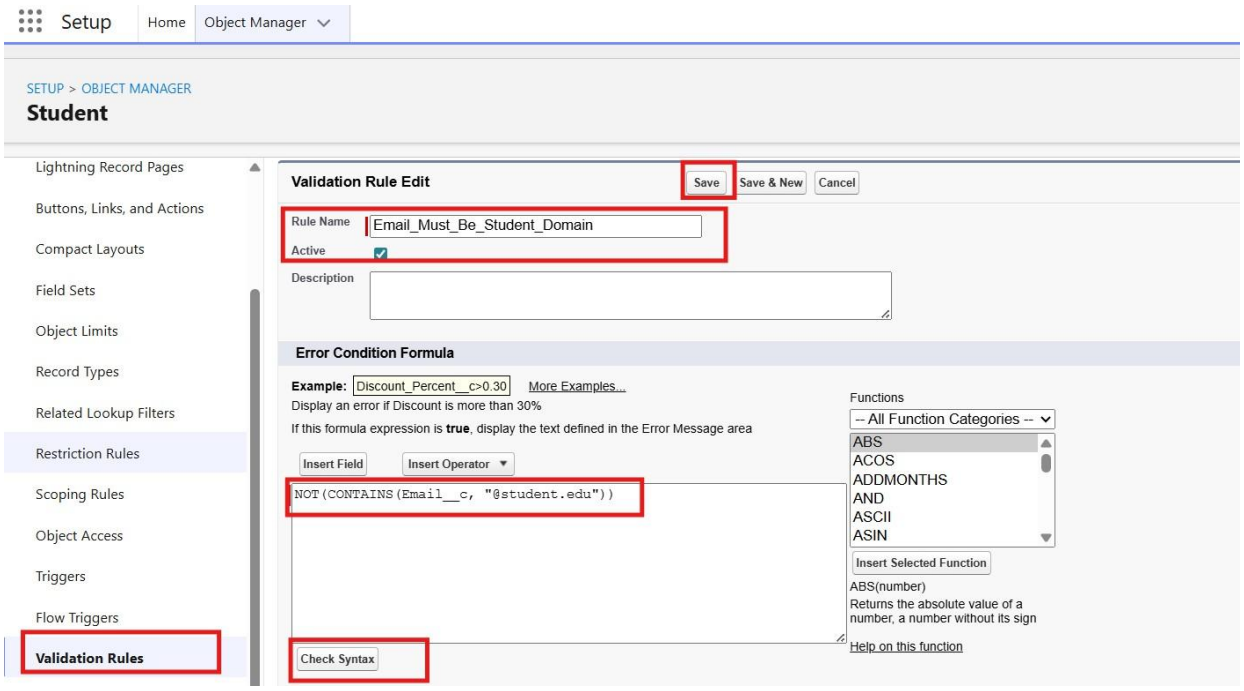
Purpose:
To ensure students use only allowed institutional email addresses.

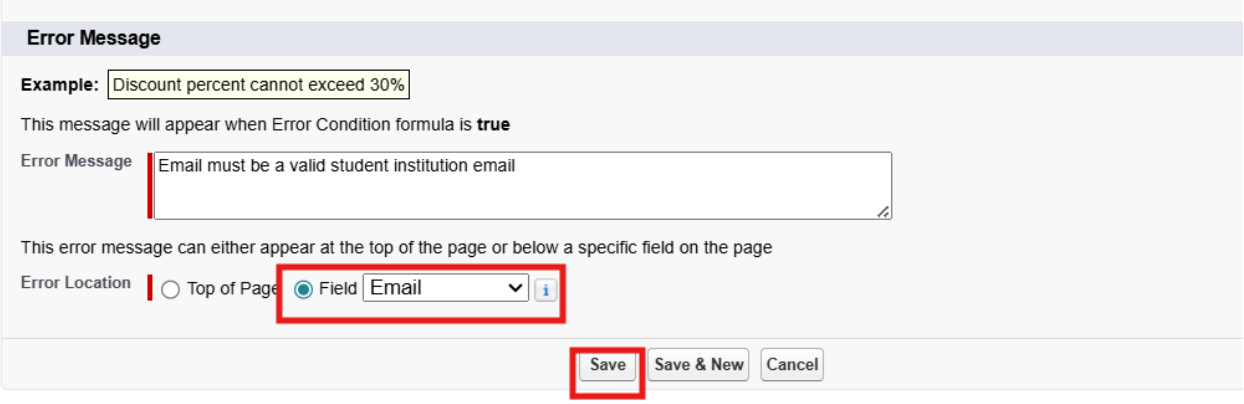
Steps:

Go to Setup → Object Manager → Student_____c.
Click Validation Rules → New.
Rule Name: Student_Email_Domain_Validation. Error
Condition Formula: (example for @student.edu)

- NOT(CONTAINS(Email_____c, "@student.edu"))

Error Message: Email must be a valid student institutional email. Error Location:
Field → Email_____c.
Click Save.






## Activity 2: Fees Must Be Paid Before Approval

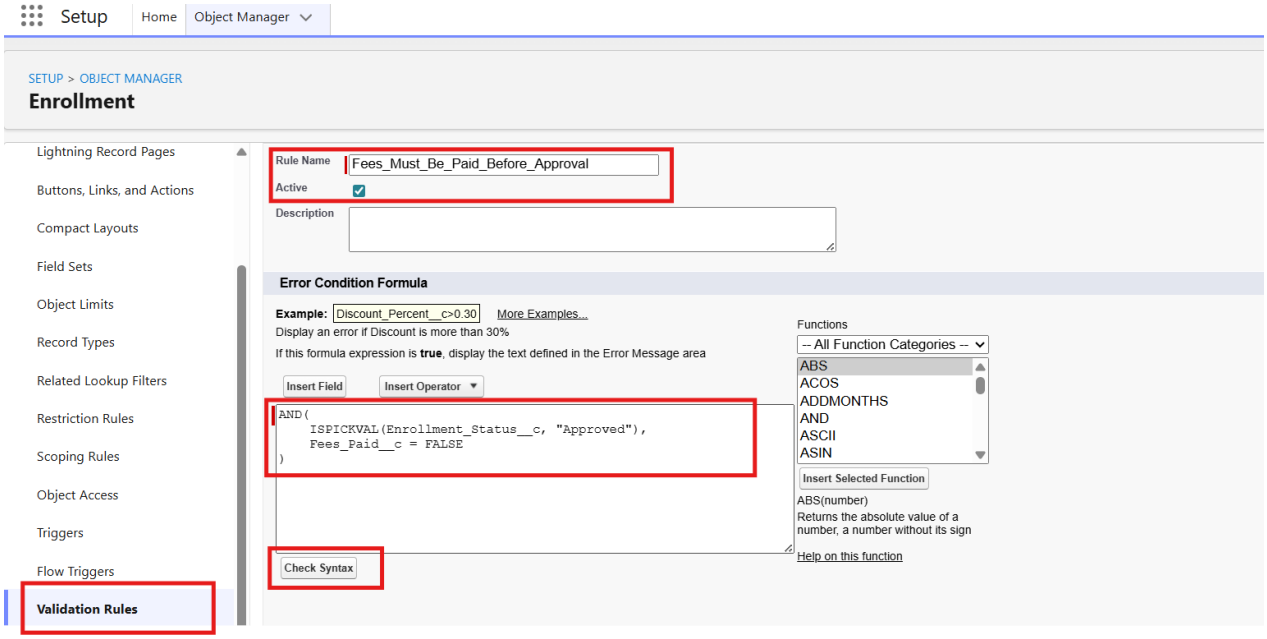
Purpose:
Prevent Enrollment from being approved until fees are paid.

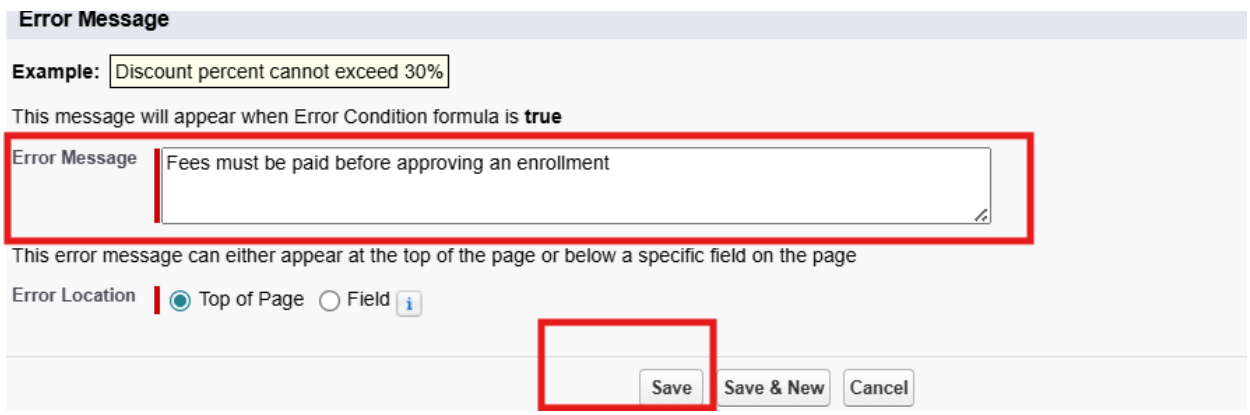
Steps:

1. Go to Setup → Object Manager → Enrollment_____c.
2. Click Validation Rules → New.
3. Rule Name: Fees_Must_Be_Paid_Before_Approval.
4. Error Condition Formula:

```
AND(
  ISPICKVAL(Enrollment_Status_____c, "Approved"),
  Fees_Paid_____c = FALSE
)
```

5. Error Message: Fees must be paid before approving an enrollment.
6. Error Location: Top of Page (or field as needed). Click Save.



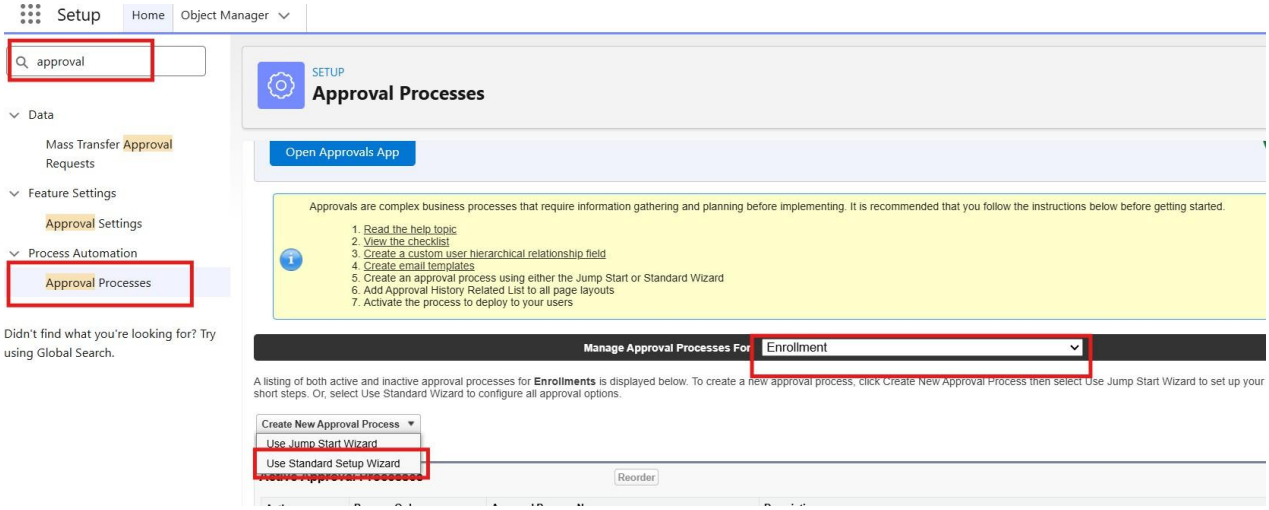

## Milestone 6 – Approval Process

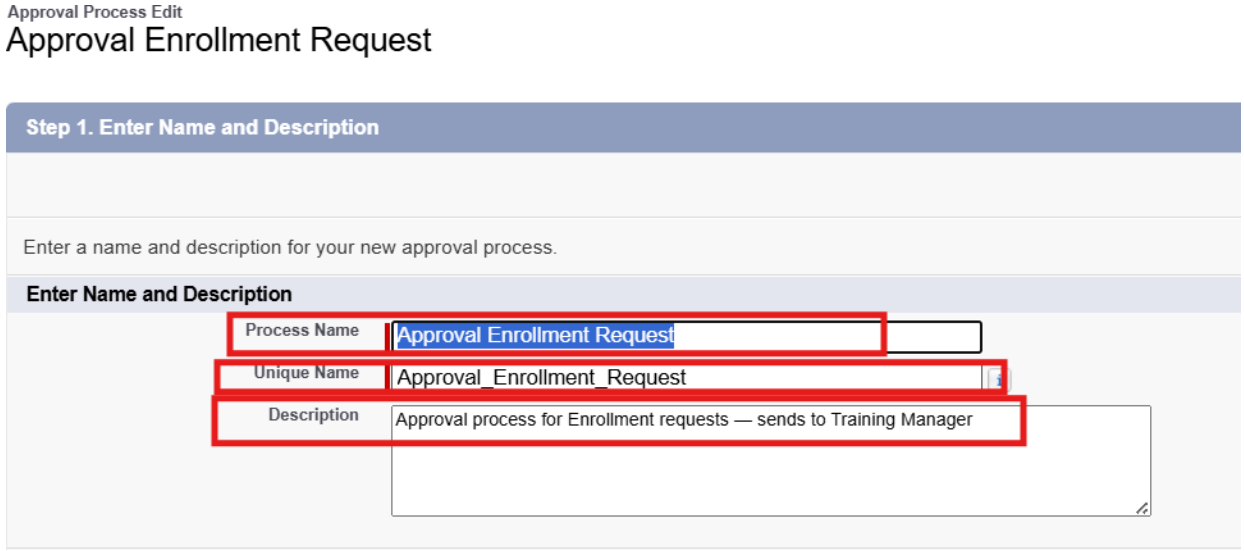
## Activity 1: Create Approval Process for Enrollment Requests

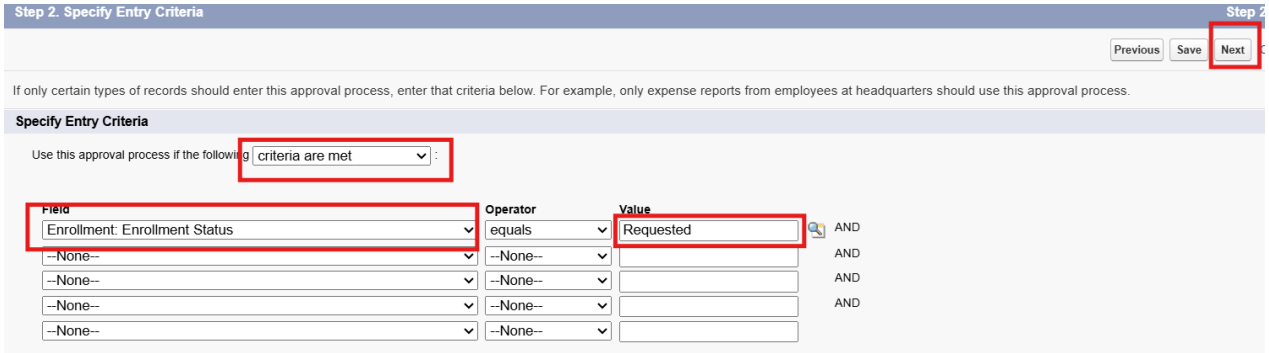
Purpose:
Route Enrollment requests to Training Manager for approval or rejection.

Steps:

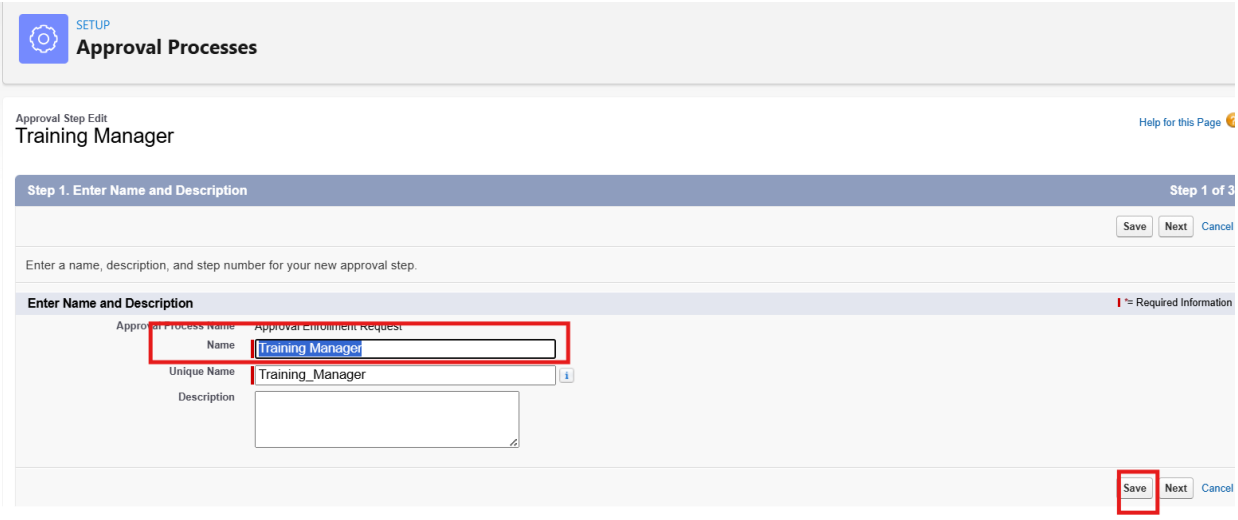
1. Go to Setup → Approval Processes → select Enrollment_____c.
2. Click Create New Approval Process → Use Standard Setup.
3. Process Name: Approval_Enrollment_Request.
4. Entry Criteria: Enrollment_Status_____c = "Requested".
5. Save and configure steps.



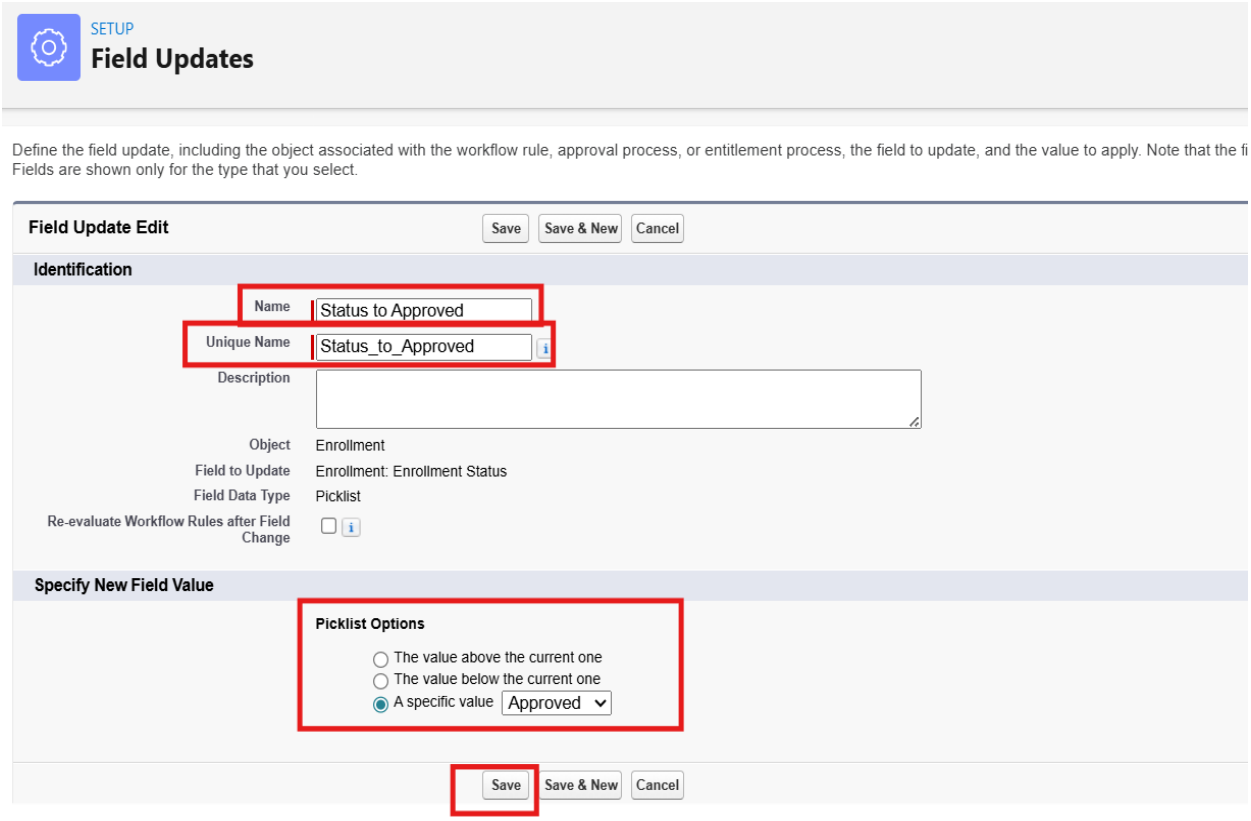





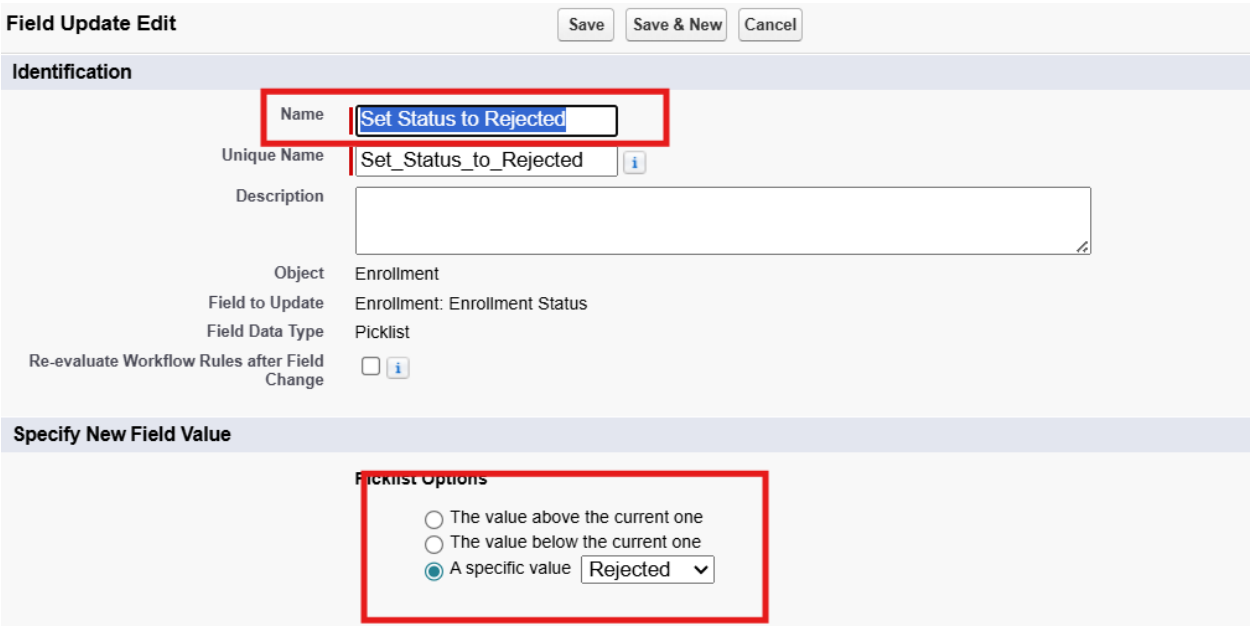

6. Add Approval Step:
7. Approver: specific Training Manager user.



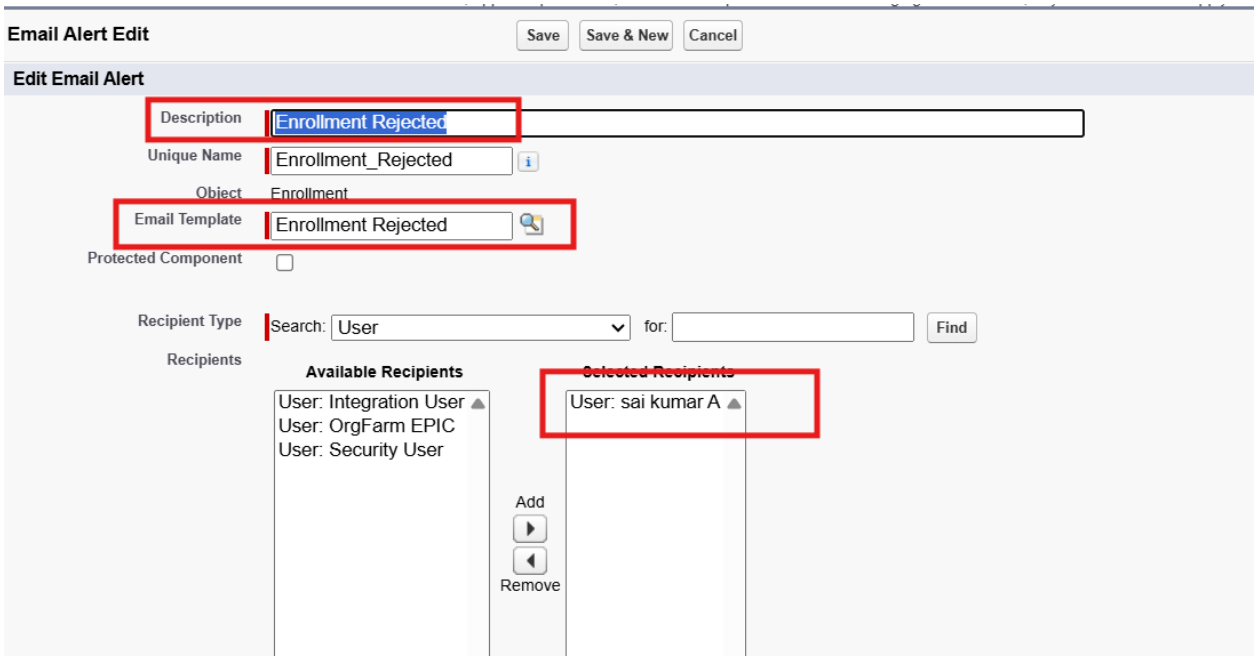

8. Under Final Approval Actions:
9. Add Field Update to set Enrollment_Status_____c = "Approved".

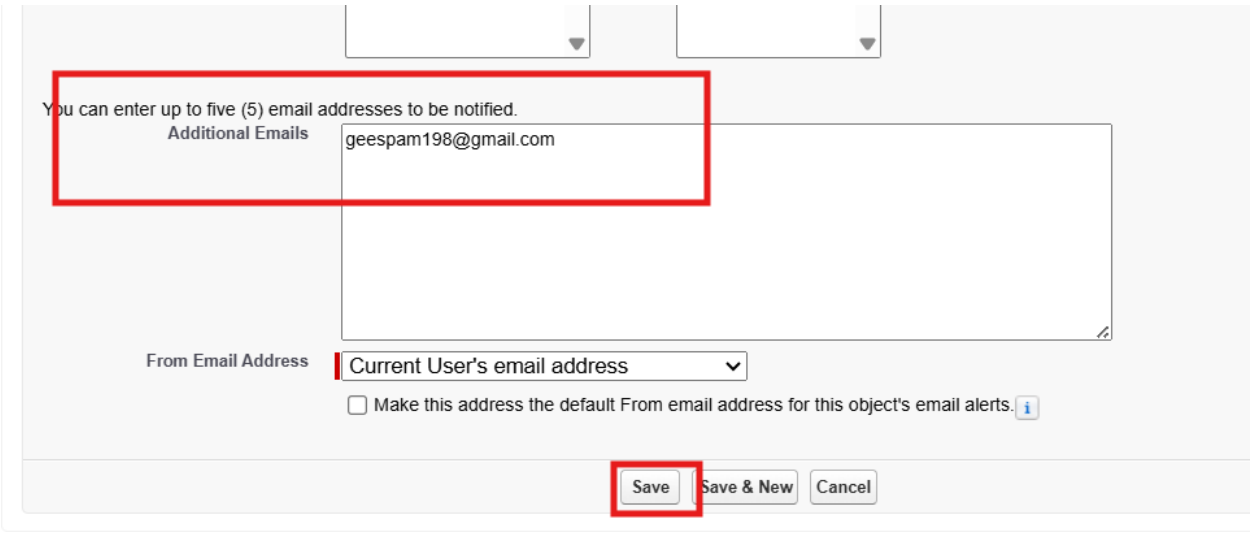



10. Under Final Rejection Actions:
11. Add Field Update to set Enrollment_Status_____c = "Rejected".




12. Add Email Alert to send rejection email to student.




13. Activate the approval process.

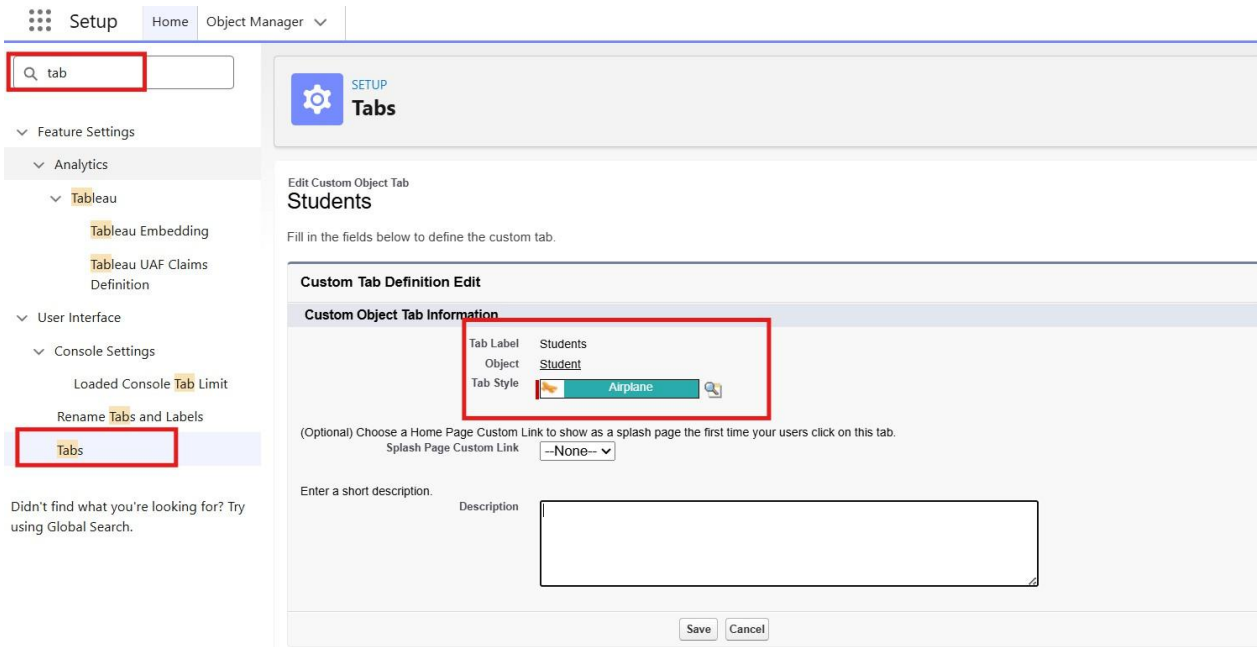
## Milestone 7- Tabs

## Activity 1: Creating a Tab for the Student Object:

Go to Setup.

1. In the Quick Find box, type Tabs.
2. Under Custom Object Tabs, click New.
3. Select Student from the object list.

4. Choose a tab style (icon).
5. Click Next, assign it to all profiles, and then click Save.




## Activity 2: Creating Remaining Tabs

Now create the Tabs for the remaining Objects, they are "Courses, Instructors, Enrollment". Follow the same steps as mentioned in Activity -1

## Milestone 8 - The Lightning App

## Activity 1: Creating the Lightning Application for the Project

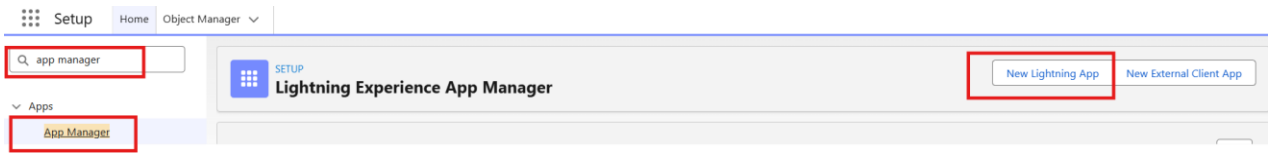
Navigate to Setup.

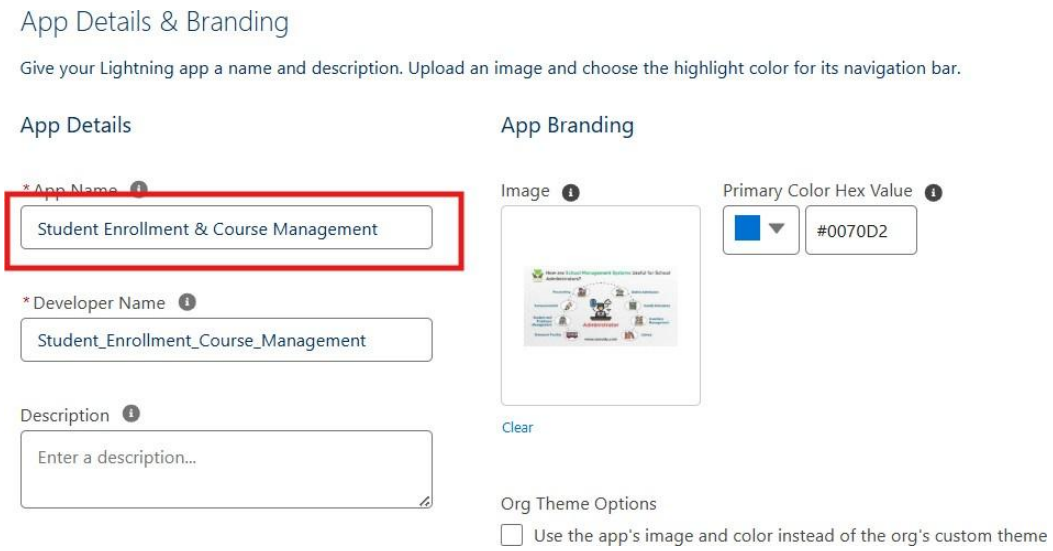
Enter App Manager in the Quick Find search box. Click New Lightning App.

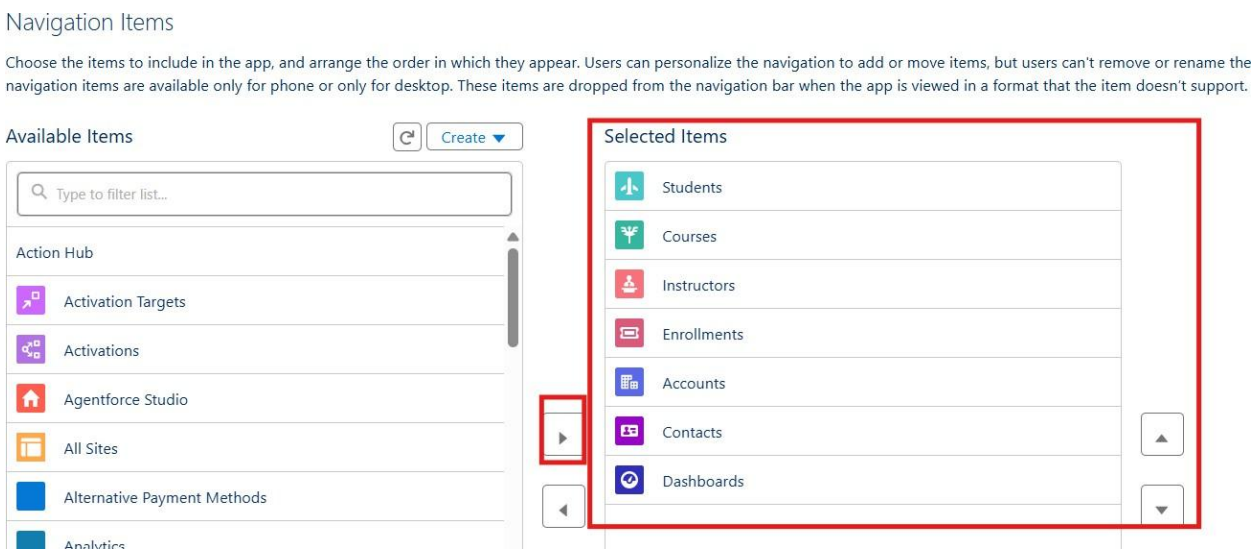
Provide the app details:
App Name: Student Enrollment & Course Management Developer Name auto-fills
Description: Optional, but can describe the purpose of the app Click Next and choose the following settings:
App Branding (optional): Add logo or select color theme Navigation Style: Choose Standard Navigation
Click Next and assign the app to specific profiles (e.g., **Admin, Enrollment Officer, Instructor**).
Click Next to select the items (tabs) to be included in the app. Add:

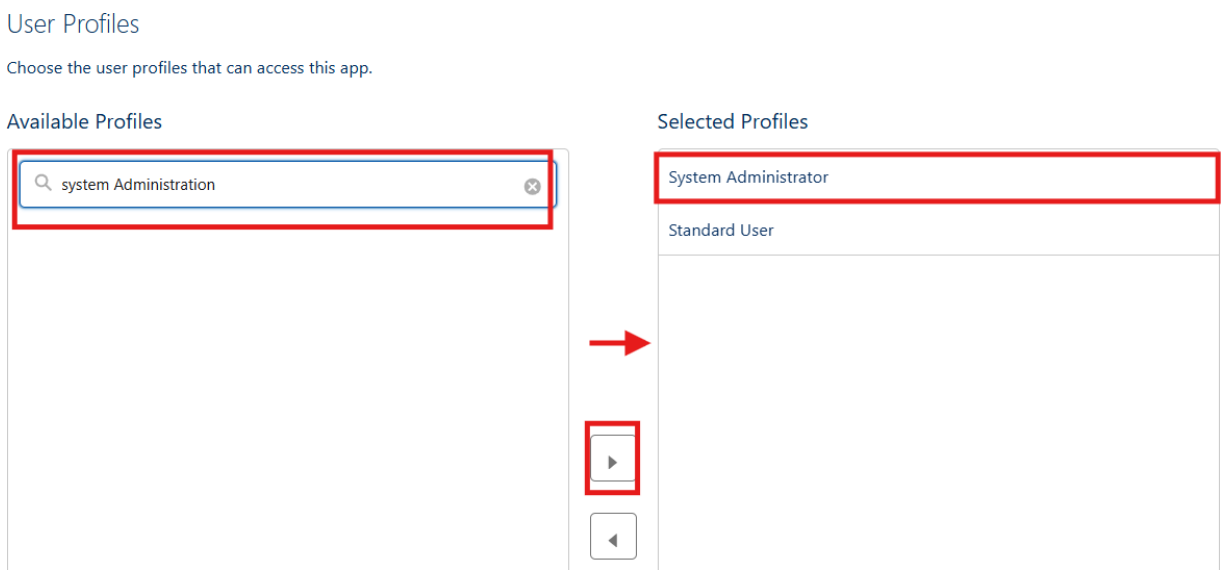
1. Student
2. Course
3. Instructor
4. Enrollment
5. Reports
6. Dashboards
7. Home
8. Tasks

Click Next and review the app settings. Finally, click Save & Finish.






## Milestone 9 – Page Layouts and Record:

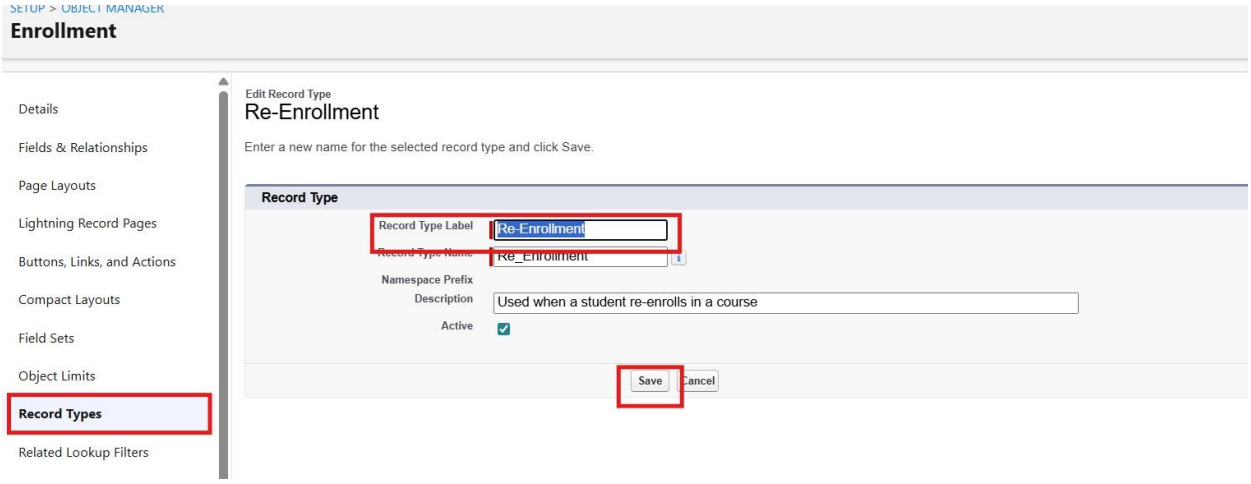
## Activity 1: Create Record Types for Enrollment:

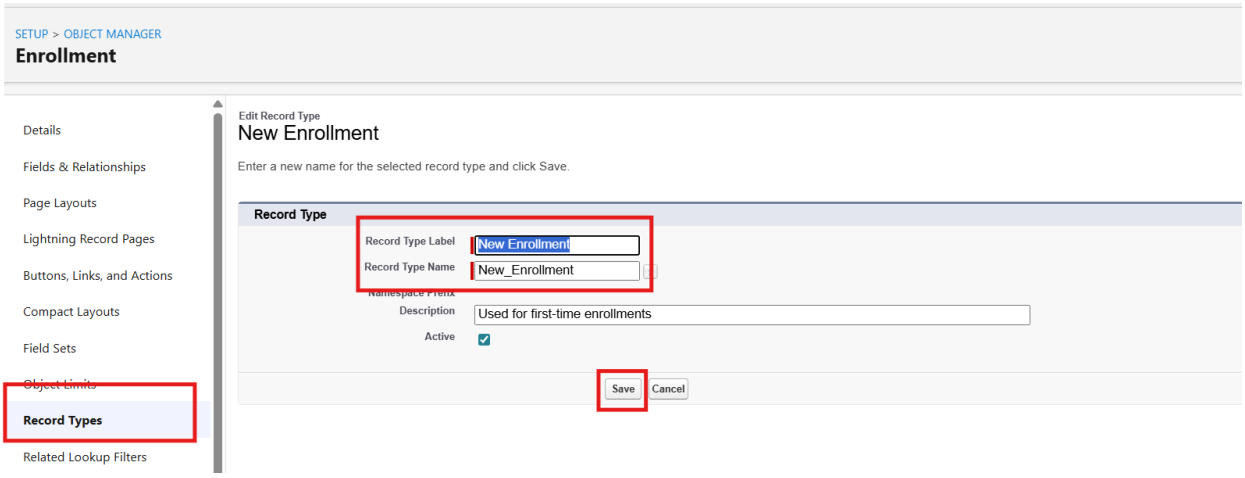
### Purpose:

To separate New Enrollment vs Re-Enrollment processes. Steps:

1. Go to Setup → Object Manager → Enrollment_____c.

2. Click Record Types → New.
3. For the first record type:
   - Existing Record Type: Master
   - Record Type Label: New Enrollment
   - Enable for required profiles (Admin, Enrollment Officer).
   - Select the default Page Layout → Save & New.
4. For the second record type:
   - Record Type Label: Re-Enrollment
   - Enable for same profiles.
   - Select the default Page Layout → Save.



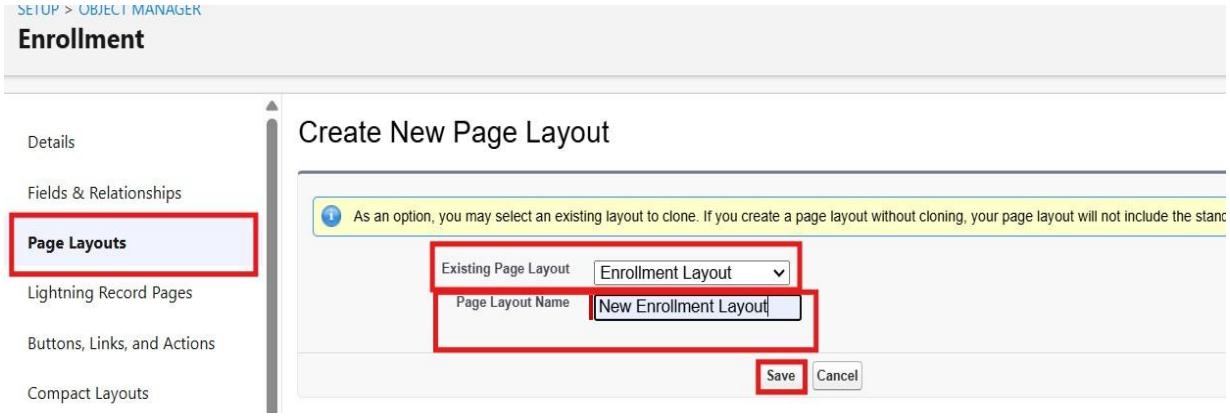

## Activity 2: Separate Page Layo Createuts for Each Enrollment Record Type

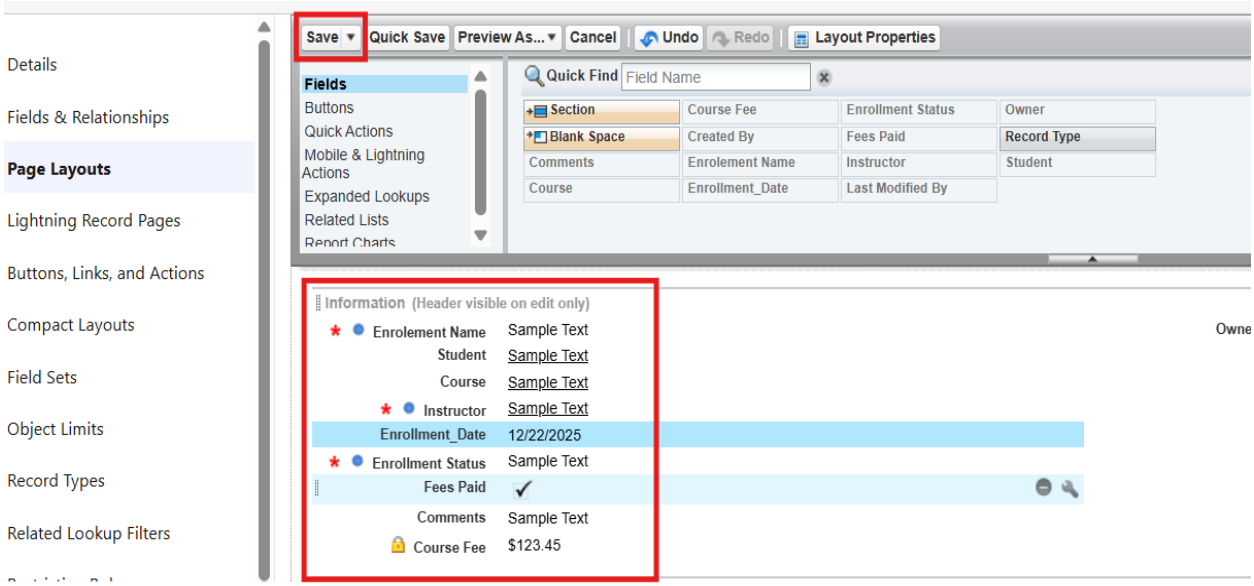
## Purpose:

To show different fields for New Enrollment vs Re-Enrollment.

## Steps:

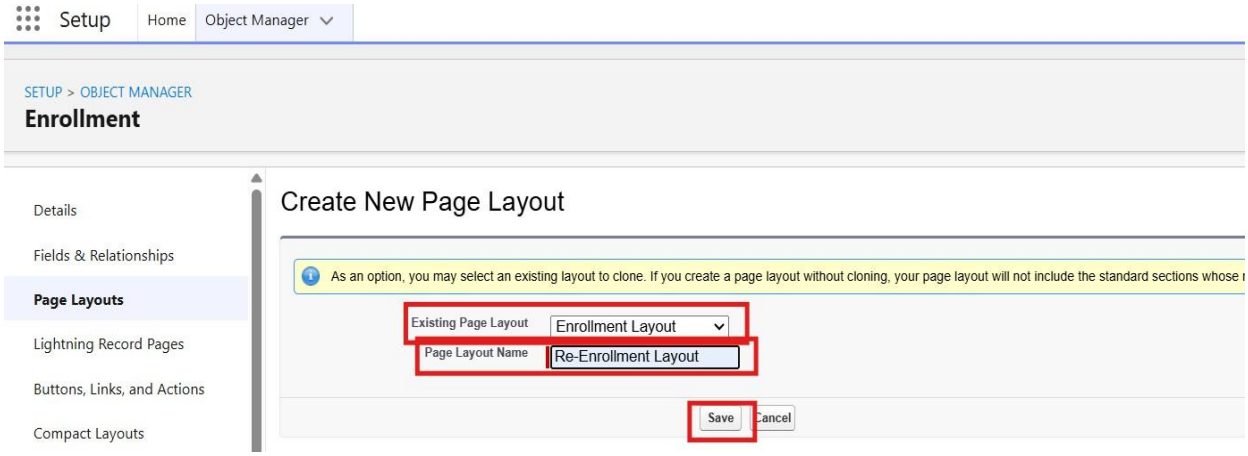
1. Enrollment_____c object → click Page Layouts.
2. Click New to create a layout for New Enrollment:
   - Clone from existing Enrollment Layout → name it New Enrollment Layout → Save.
   - Keep fields relevant for first-time enrollment: Student, Course, Instructor, Enrollment Date, Status, Fees, etc.
   - Remove unnecessary re-enrollment fields if any.
   - Click Save.

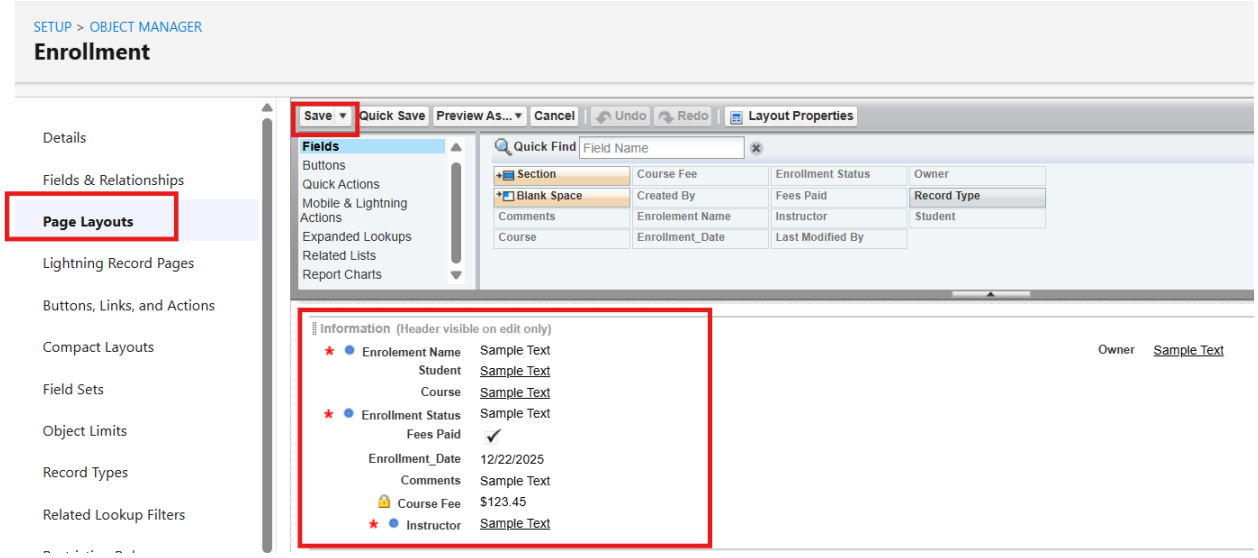



3. Again, click New to create Re-Enrollment Layout:

- Clone existing layout → name it Re-Enrollment Layout → Save.
- Include fields like previous Enrollment reference (if any), Re-Enrollment Reason, Status, Fees, etc.
- Click Save.




4. Click Page Layout Assignment → Edit Assignment.
5. For each profile and record type:

- New Enrollment → use New Enrollment Layout. 6.Re-Enrollment → use Re-Enrollment Layout.

Click Save.

## Activity 3: Enrollment R Addelated Lists to Parent Objects:

Purpose:

To see all Enrollments from Student, Course, and Instructor records.

a) Student → Enrollment Related List

Go to Setup → Object Manager → Student_____c.

Click Page Layouts → open the main layout.

In the palette, select Related Lists.

Drag Enrollments related list onto the layout. Customize columns if needed → Save.

b) Course → Enrollment Related List

Go to Object Manager → Course_____c → Page Layouts. Open layout → add Enrollments related list.

Save.

c) Instructor → Enrollment Related List

Go to Object Manager → Instructor_____c → Page Layouts.

Open layout → add Enrollments related list.

Save.

## Milestone 10 – Automation with Flow

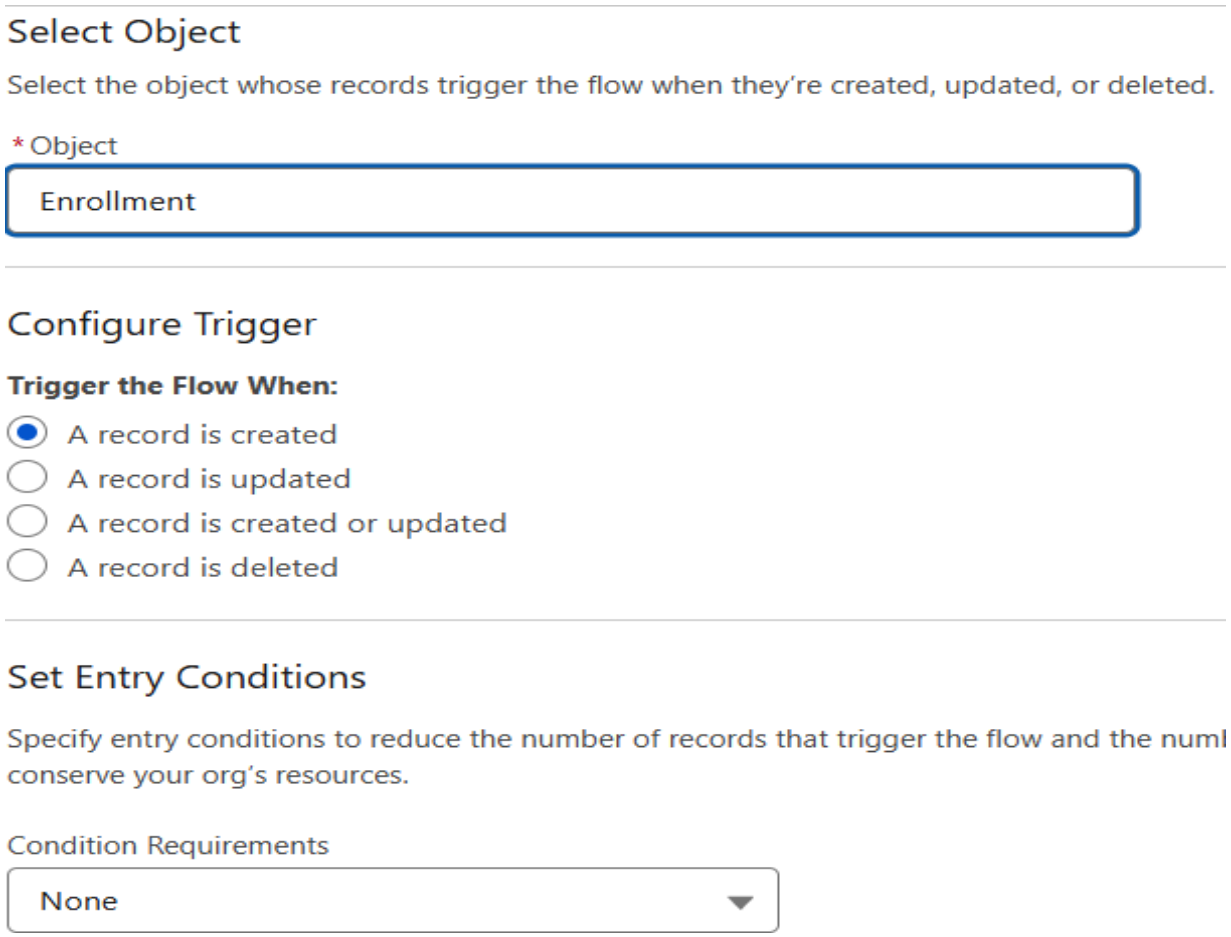
## Activity 1: Auto-Populate Enrollment Date (Record-Triggered Flow)

Purpose:

Set Enrollment_Date_____c to today when Enrollment is created.

Steps:

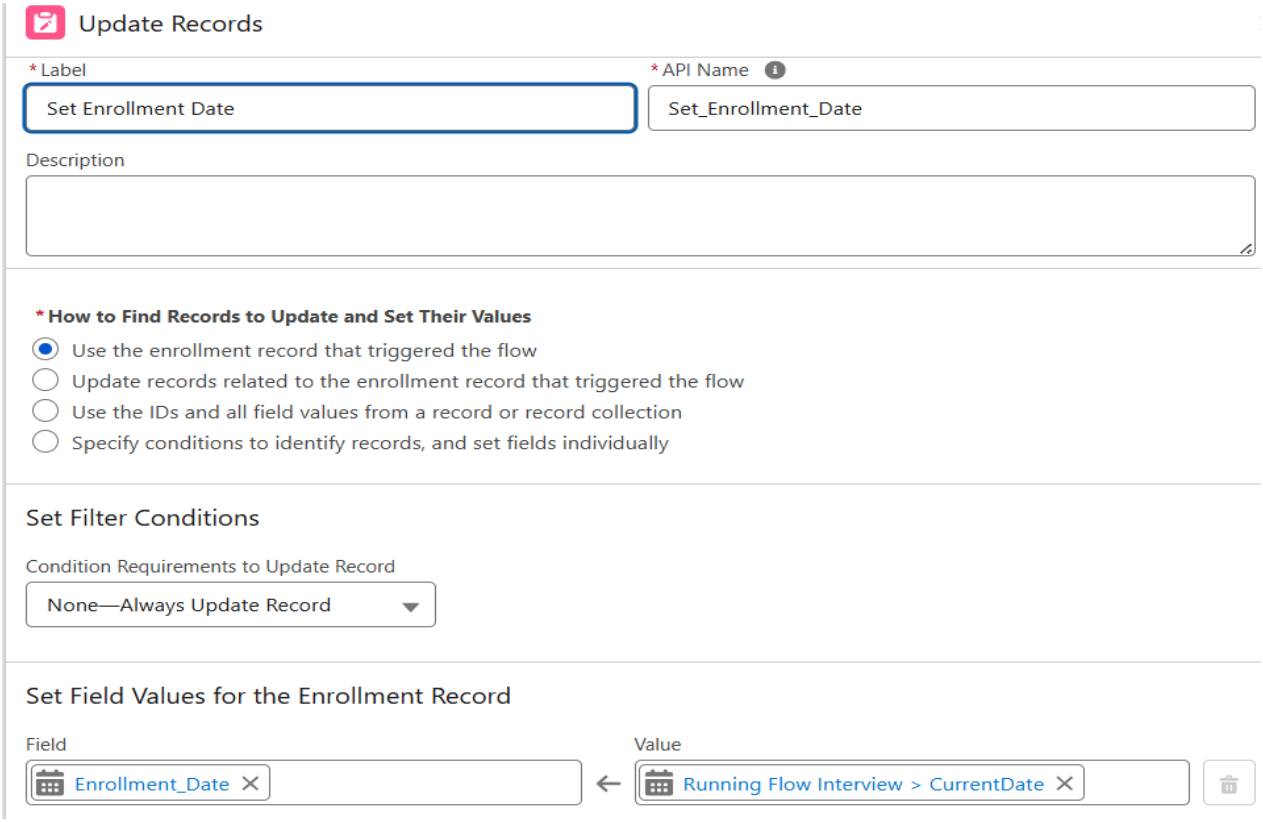
1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow → Create.
3. Object: Enrollment_____c.
4. Trigger: A record is created.
5. Optimize for Actions and Related Records → Done.




6. Add Update Records element:

- Label: Set Enrollment Date.
- Use record that triggered the flow.
- Field: Enrollment_Date_____c → Value: $Flow.CurrentDate.

7. Save as Set Enrollment Date → Activate.




## Activity 2: Send Email When Enrollment Approved

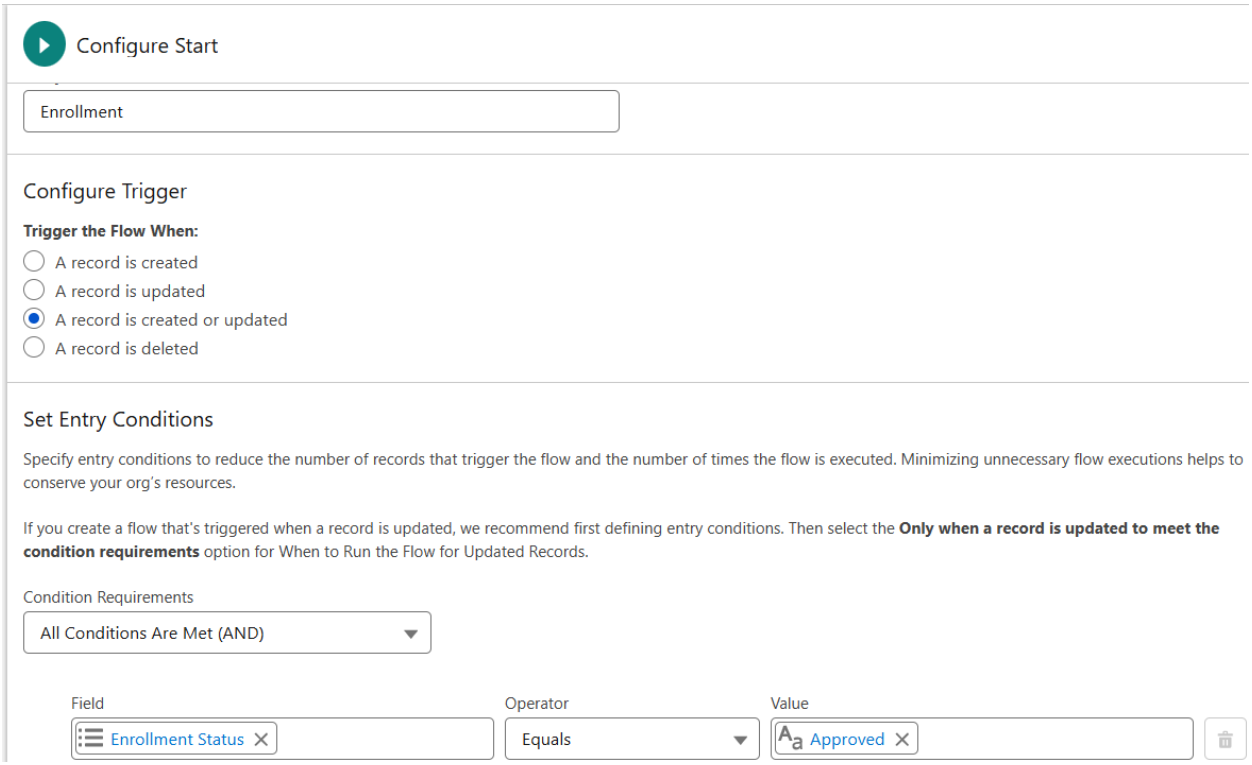
### Purpose:

When Enrollment_Status_____c = Approved, send email, alert admin, and set Student to Active.

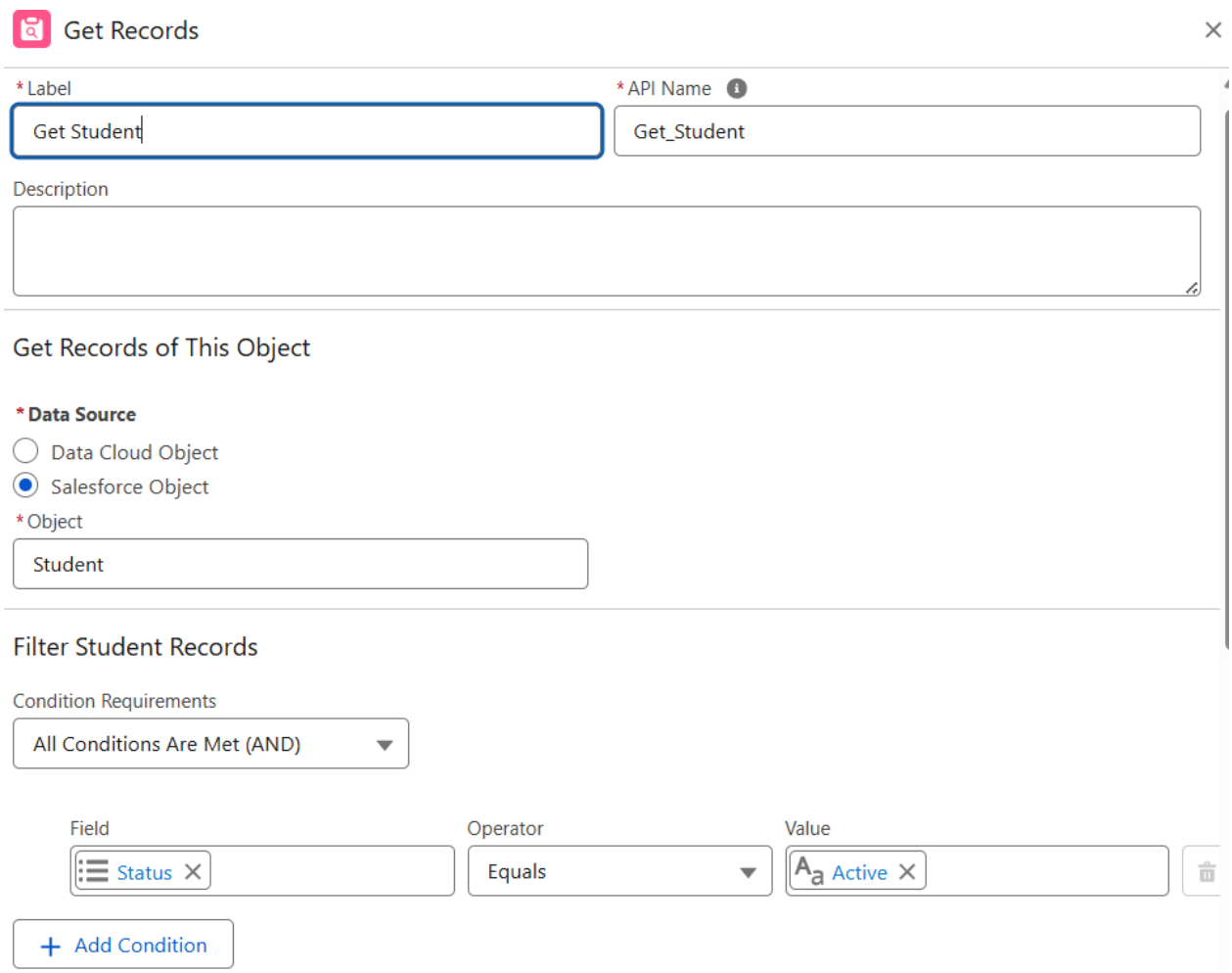
Steps (high level for document):

1. Create Email Template and Email Alert for student notification.

2. Go to Flows → New → Record-Triggered Flow.
3. Object: Enrollment_____c, Trigger on update.
4. Condition: Enrollment_Status_____c Equals "Approved".




5. Add Get Records to fetch related Student_____c.




6. Add Action (Email Alert) to send email to student.

Enrollment Approved EmailAlert

* Label

Send Student Email

* API Name

Send_Student_Email

Description

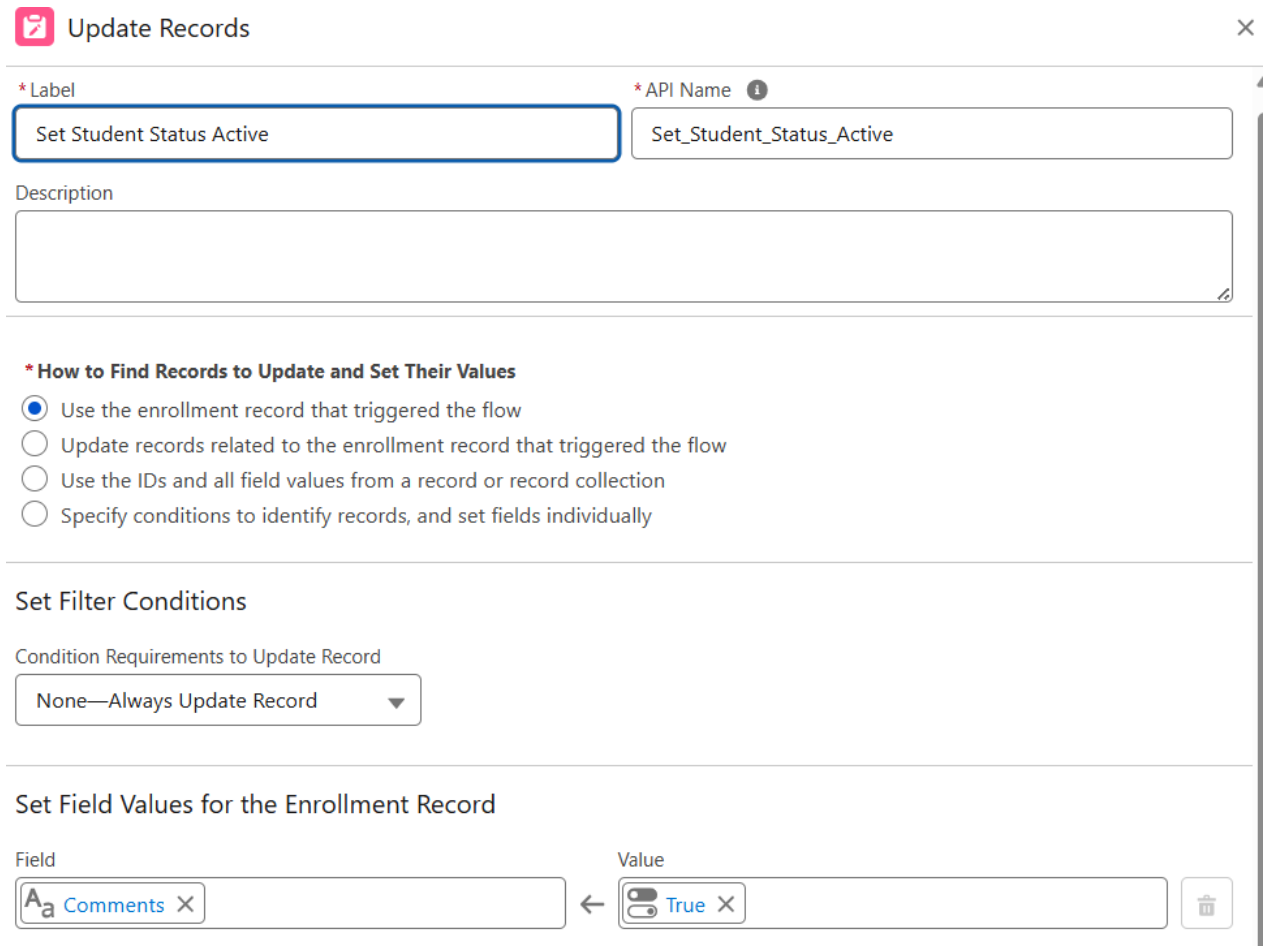
Use values from earlier in the flow to set the inputs for the "Enrollment Approved EmailAlert" email alert. To use its outputs later in the flow, store them in variables.

Set Input Values

* Record ID

Triggering Enrolement__c > Record ID

7. Add Action to notify Admin if needed.
8. Add Update Records to set Student_____c.Status_____c = "Active".




9. Save as Enrollment Approved Action → Activate.

## Activity 3: Auto-Assign Instructor Based on Course Category

### Purpose:

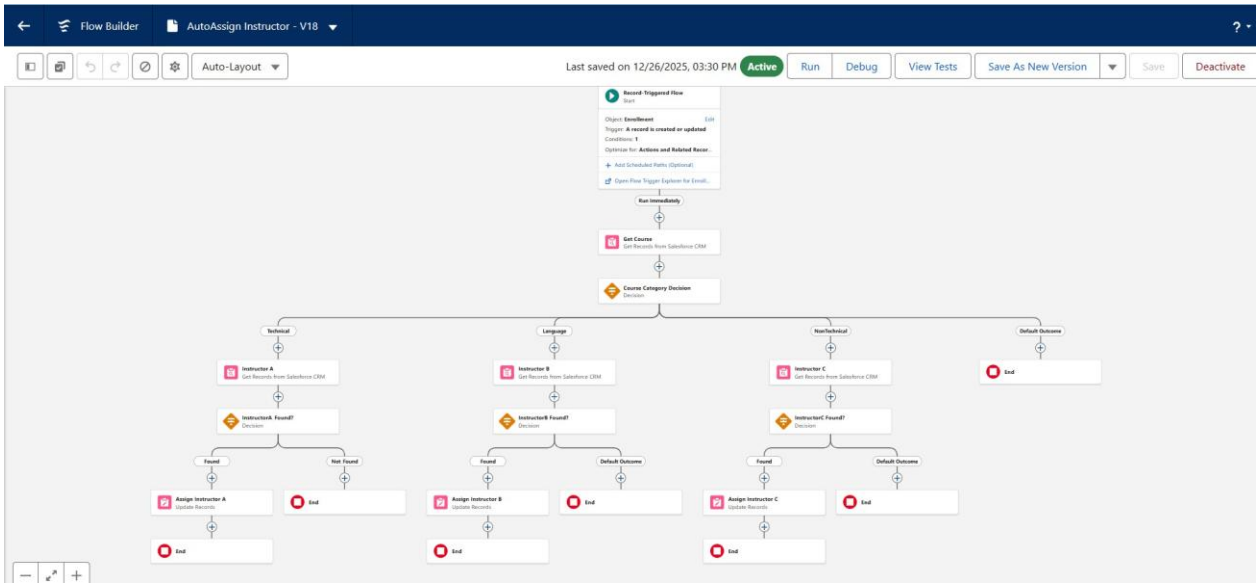
Assign Instructor automatically based on Course category.

### Rules:

Technical → Instructor A
Language → Instructor B
Non-Technical → Instructor C

Steps:

1. Prepare Instructor_____c records with codes (e.g., INSTR_A, INSTR_B, INSTR_C).
2. Create Record-Triggered Flow on Enrollment_____c (created/updated).
3. Add Get Records to fetch related Course_____c.
4. Add Decision element on Course_____r.Category_____c:
5. Technical / Language / Non-Technical outcomes.
6. For each outcome:
7. Get Instructor record (A/B/C) based on code or name.
8. Update Records: set Instructor_____c = Get_Instructor_X.Id.
9. Save as AutoAssign_Instructor → Activate.



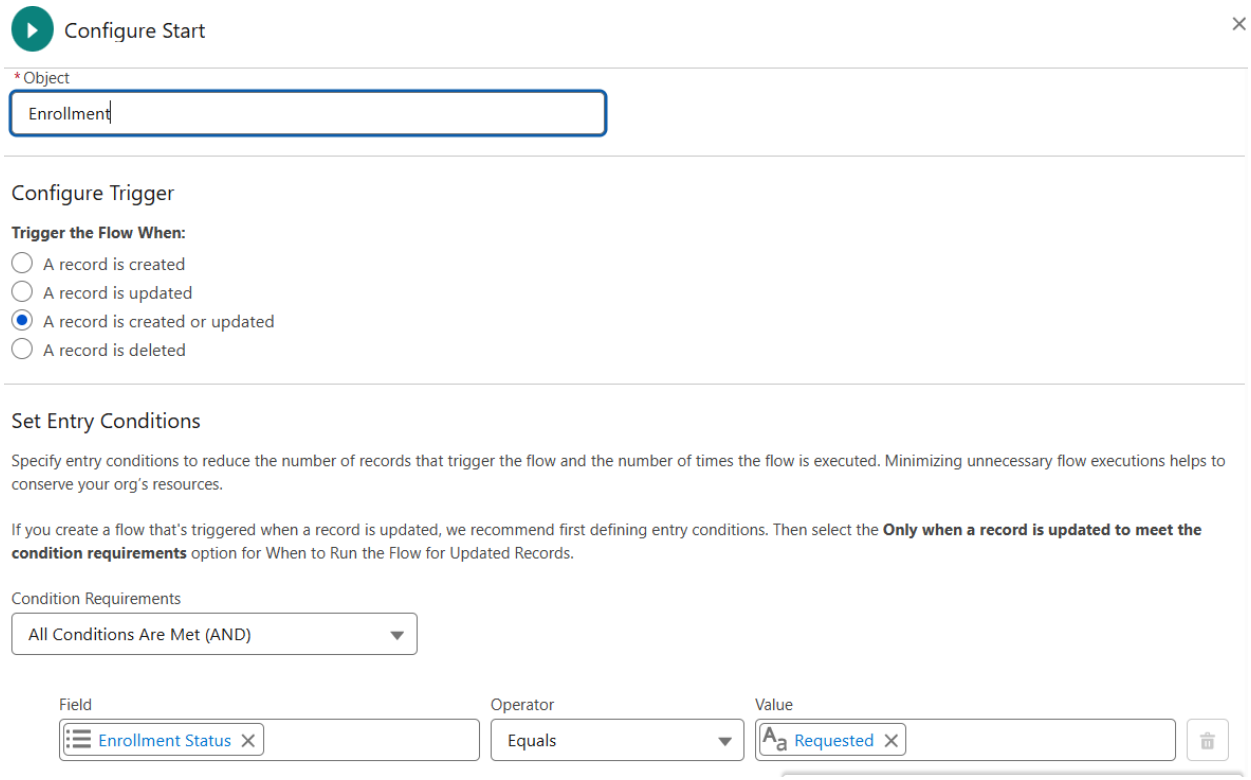

## Activity 4: Auto-Create Task for Follow-Up

Purpose:

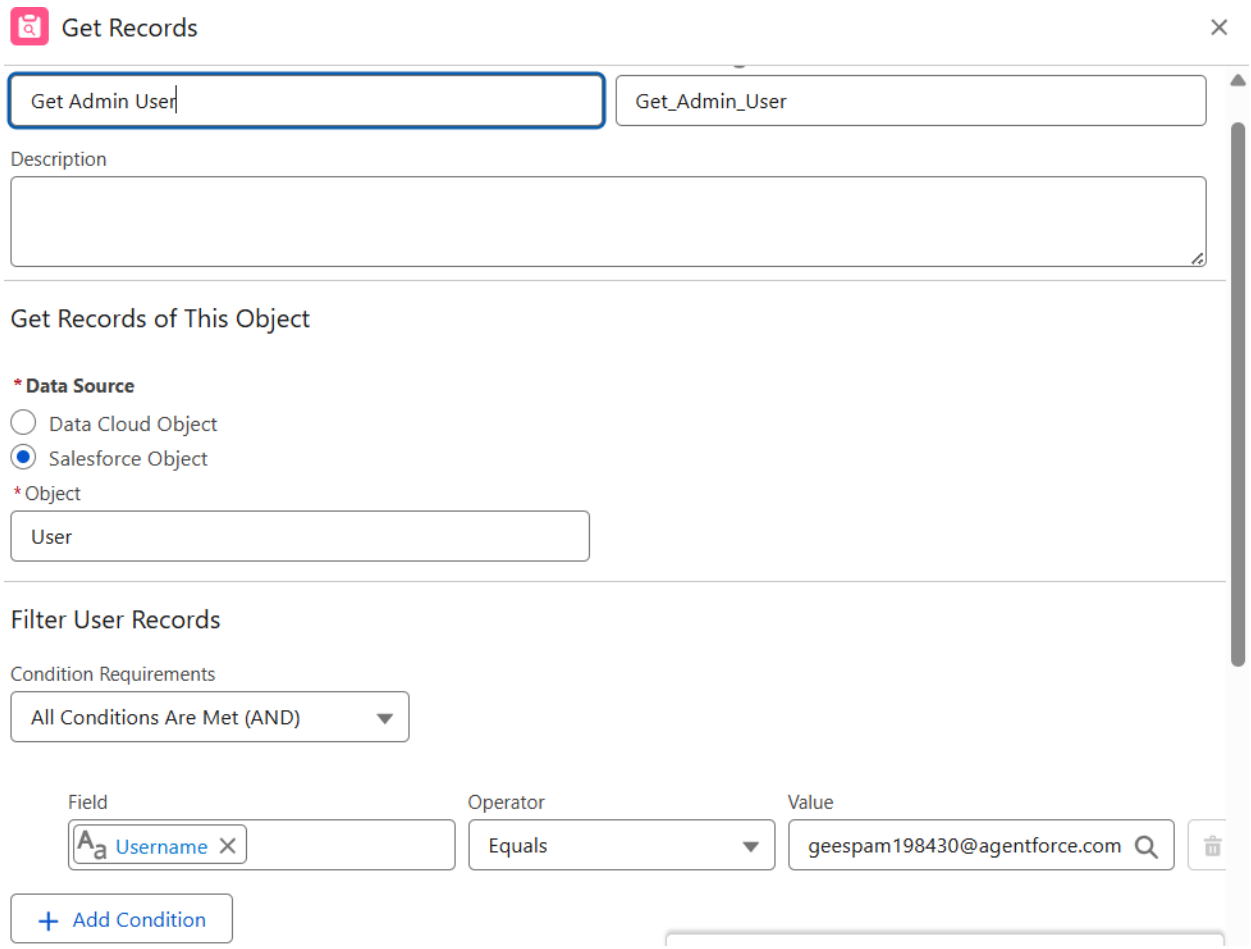
When Enrollment is “Requested”, create a follow-up task for Admin.

Steps:

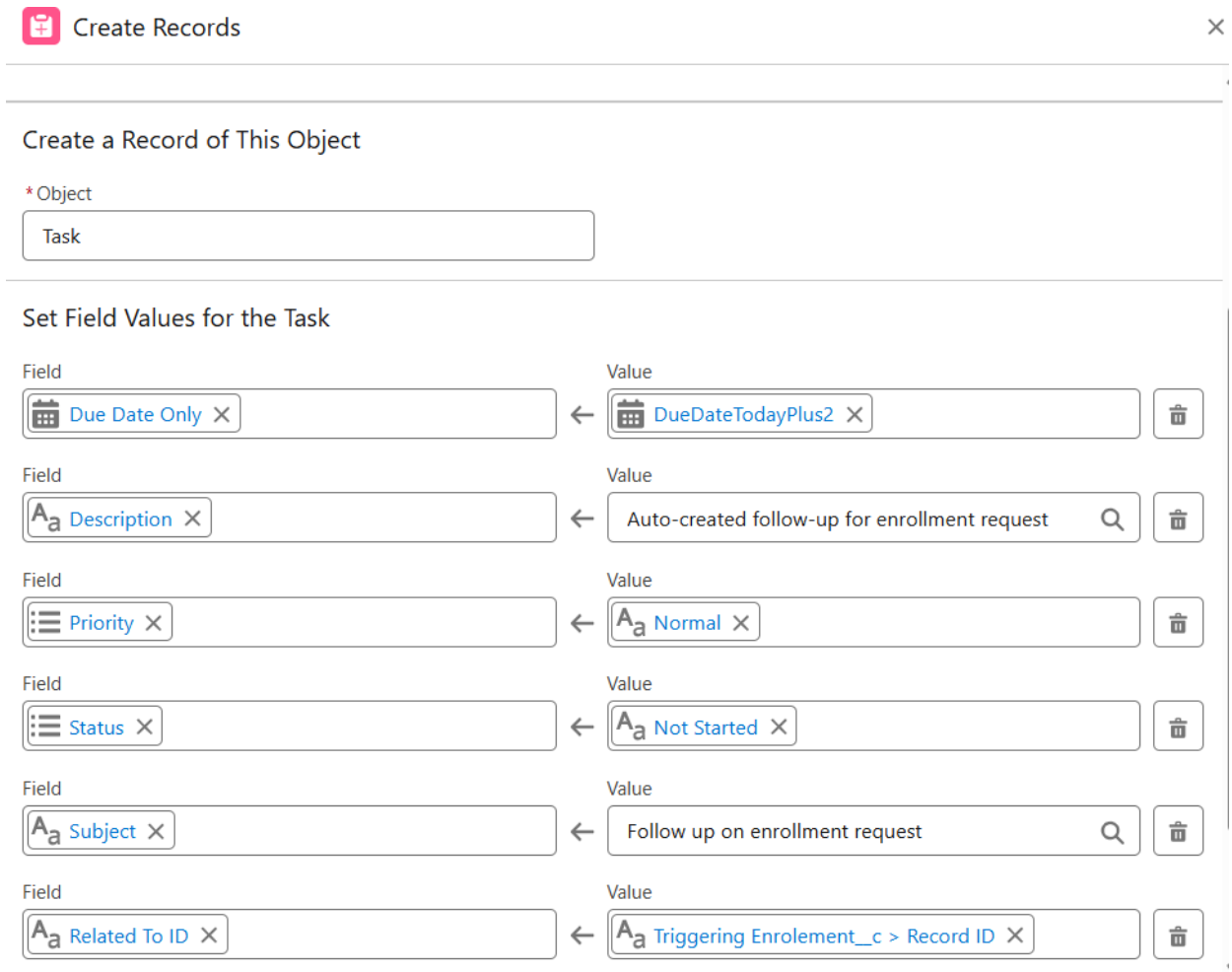
1. Create Record-Triggered Flow on Enrollment_____c.
2. Trigger: Record created or updated; condition Enrollment_Status_____c = "Requested".




3. Add Get Records to fetch Admin user (by Username / Profile).
4. Add Formula (Date): TODAY() + 2 for due date.




5. Add Create Records for Task:
6. Subject: Follow up on enrollment request.
7. OwnerId: Admin user Id.
8. WhatId: Enrollment Id.
9. ActivityDate: formula Today + 2.

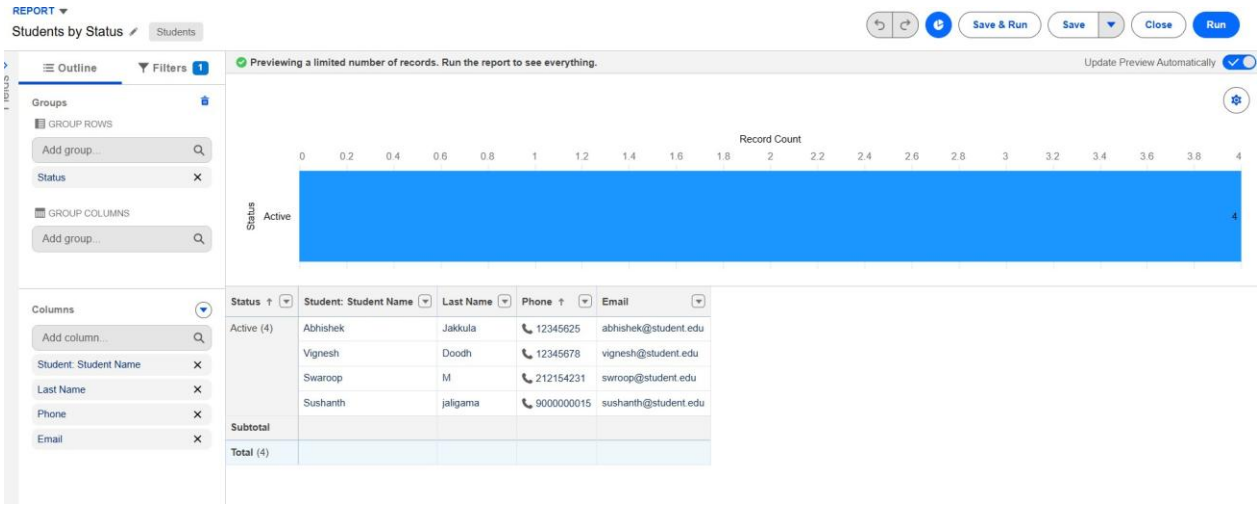



10. Save as Create_Followup_Task → Activate.

## Milestone 11 – Reports:

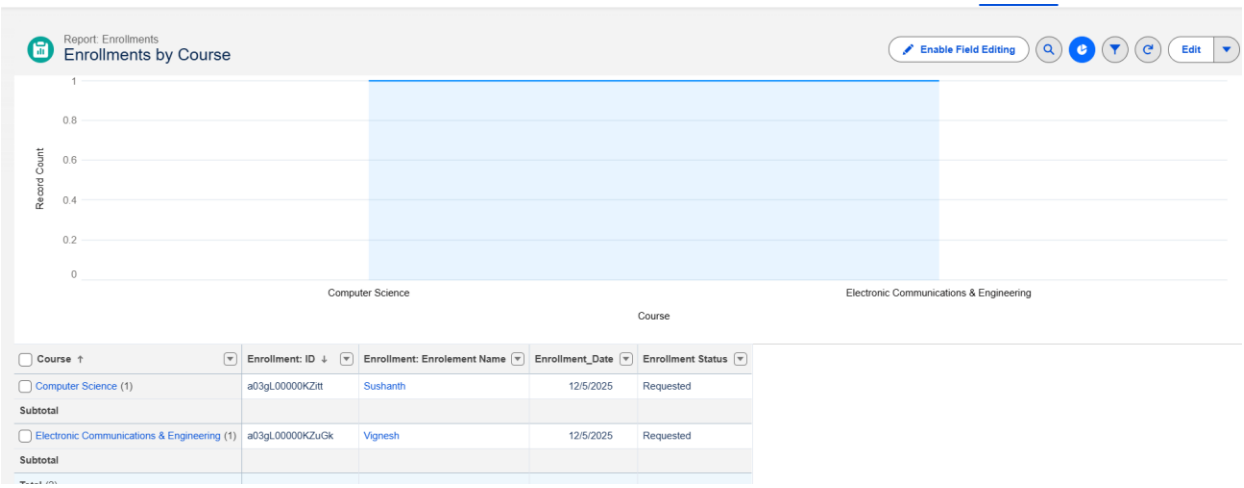
## Activity 1: Create "Students by Status" Report

1. Go to **Reports → New Report**.
2. Report Type: Students.
3. Group rows by Status_____c.
4. Add columns as needed (Name, Email, Phone).
5. Save as Students by Status.



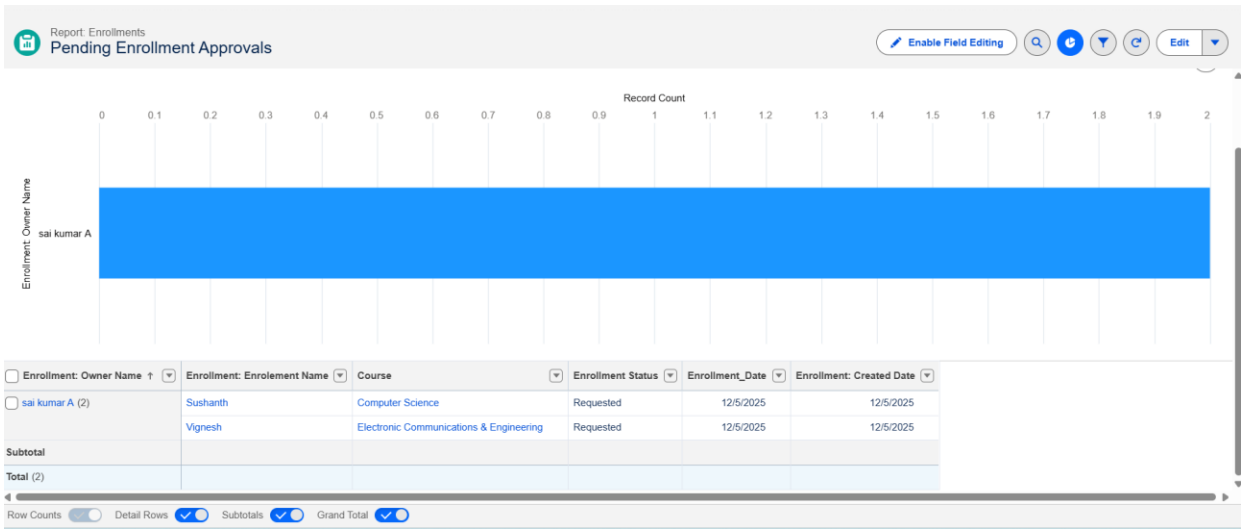

## Activity 2: Create "Enrollments by Course" Report

1. New Report → Report Type: Enrollments.
2. Grouprows by Course_____r.Name.
3. Show Record  Count per course.
4. Save as Enrollments by Course



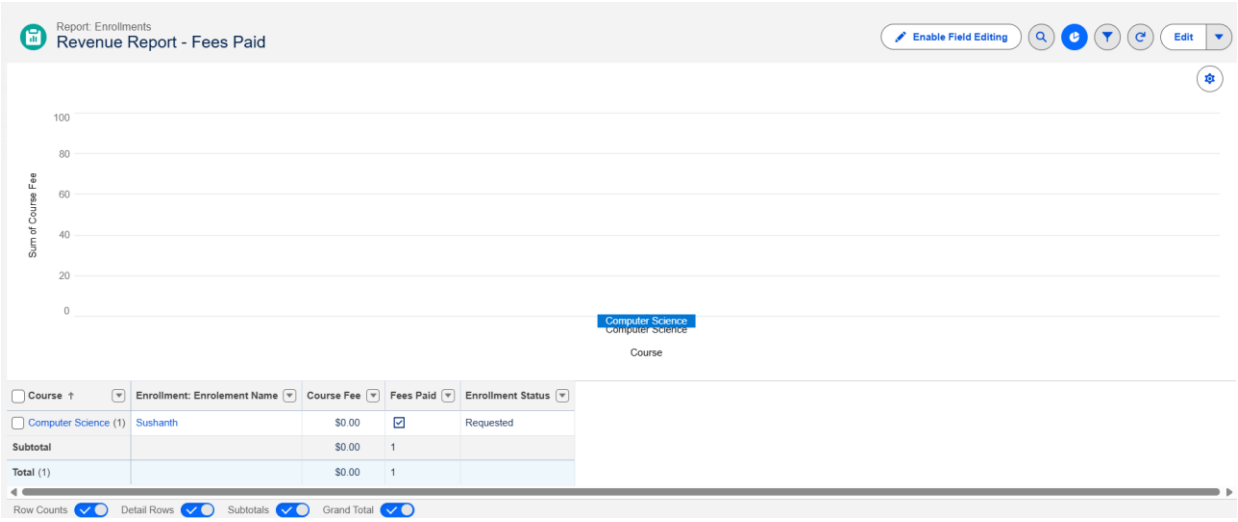
## Activity 3: Create “Pending Enrollment Approvals” Report

1. New Report → Enrollments report type.
2. Filter: Enrollment_Status c = "Requested".
3. Add columns (Student, Course, Owner, Created Date).
4. Save as Pending Enrollment Approvals.




## Activity 4: Create “Revenue Report (Fees Paid)” Report

1. New Report → Enrollments report type.
2. Filter: Fees_Paid_____c = TRUE.
3. Group by Course_____r.Name.
4. Summarize Course_Fee_____c or similar fee field.
5. Save as Revenue Report - Fees Paid.

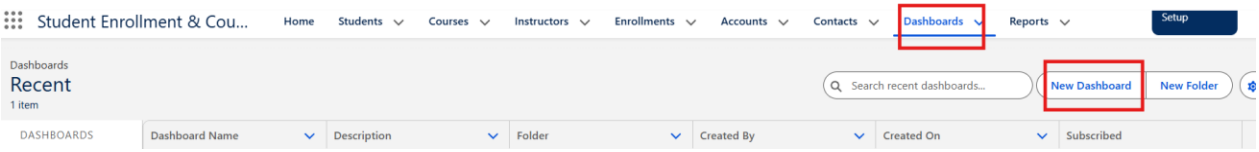



## Milestone 12 – Dashboards:

## Activity 1: Create "Student Managemet Dnashboard"

### Steps:

1. Go to **Dashboards → New Dashboard.**
2. Name: Student Management Dashboard.



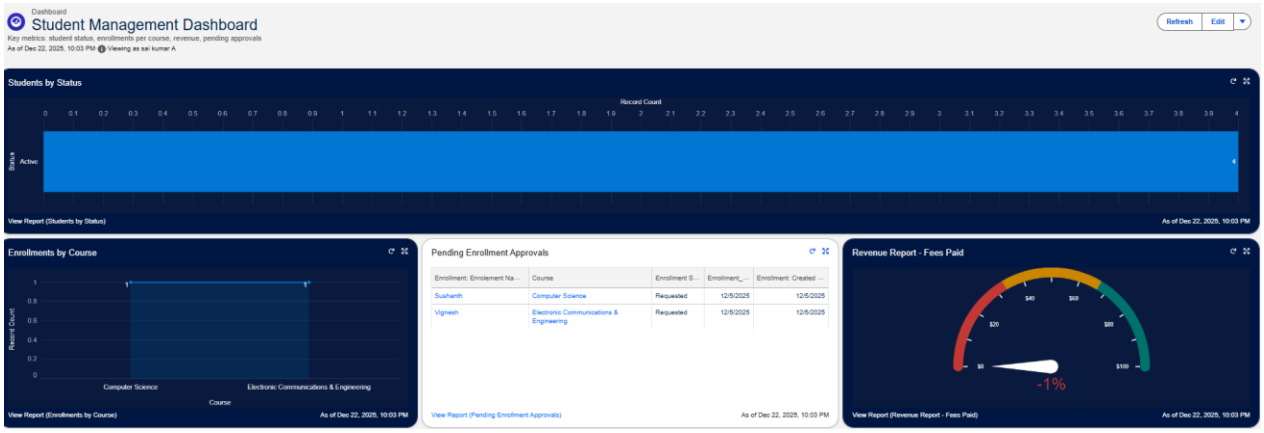

3. Select folder → **Create**.
4. Select new Dashboard folder and Save.

Add components:

- **Donut Chart** → using Students by Status report.
- **Bar Chart** → using Enrollments by Course report.

- **Gauge →** using Revenue Report – Fees Paid (sum of revenue field).
- **Table →** using Pending Enrollment Approvals report.

4. Arrange components on canvas.




Click **Save → Done.**

## Milestone 13 – Security Setup

### Activity 1: Configure Profiles

**) :**

1. Go to **Setup → Profiles → New** (or Clone**Profiles Required:**

- **Admin** – Full access.
- **Enrollment Officer** – Create/Update Enrollments.
- **Instructor** – Read-only access to assigned Enrollments.

2. **Steps (example for Enrollment Officer** fName: Enrollment Officer.
3. In **Object Settings**:
    - Enrollment_____c: Read, Create, Edit.
    - Student_____c / Course_____c: Read (and Edit if required).
4. Save.

Repeat similarly for **Instructor** profile with **Read-Only**
on Enrollments

## Activity 2: Create Permission Sets

1. Go to **Setup → Permission Sets → New**.
2. Create Custom Object Access (additional access to project objects if needed).
3. Create Dashboard Editor (Create/Customize Reports & Dashboards permissions).
4. Assign to relevant users.

## Activity 3: Configure Sharing & OWD

### Steps:

1. Go to **Setup → Sharing Settings**.
2. Set **Organization-Wide Defaults**:

- **Students (Student__c):** Public Read/Write.
- **Courses (Course__c):** Public Read/Write.
- **Enrollments (Enrollment__c):** Private.

3. Save.

For Instructor access to assigned Enrollments, optionally:

- Add a **User lookup** on Enrollment (Instructor User).
- Use a simple **Record-Triggered Flow** to set **OwnerId = Instructor User** so they can see only their own records.

## 7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce user interface to verify the proper functioning of all implemented features of the Student Enrollment & Course Management System (SECMS). These include student, course, instructor, and enrollment record creation; validation rule checks for mandatory fields, eligibility, and enrollment constraints; verification of record type behavior for different course categories; flow execution for automated enrollment approvals and seat availability checks; and Apex trigger functionality to ensure automatic confirmation and record updates. Additionally, screenshots of generated reports and dashboards were documented to validate accurate data visualization, enrollment tracking, and performance monitoring. Submit the Screenshots while testing.

## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Centralized data
- Automated workflows
- Real-time reporting

### Disadvantages

- Salesforce dependency
- Requires initial configuration effort

## 9. Conclusion:

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle—from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce's automation tools such as validation rules, flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

## 10. FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights

- Mobile access support

## 11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

# Thank You